

Plant-wide Optimization of Activated Sludge Systems Using a Physically Constrained Statistical Surrogate

Egberto F. Selerio Jr.^{a,*}

^aEngineering Research and Development for Technology, School of Engineering, University of San Carlos, Philippines

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ABSTRACT

Operational optimization of activated sludge plants requires repeated solutions of coupled biological, recycle, and settling equations. Statistical surrogates can reduce this computational cost, but their predictions must respect the physical connections among treatment units. This study develops a plant-wide, physically constrained statistical surrogate that combines second-order ridge regression with joint quadratic projection. The model predicts mixer and reactor concentrations, clarifier outlet component flows, and stored solids. Projection enforces the specified component and inventory balances, separation requirements, inventory bounds, and non-negativity. A simulated plant with five reactors and a ten-layer clarifier provides 7,386 accepted development states and 1,845 accepted holdout states. Projection reduces aggregate normalized root mean square error by 8.19% and normalized mean absolute error by 15.65%. Root mean square error improves in 30 of the 32 combinations of treatment location and water-quality measure. Surrogate-based operating searches are compared with direct smooth mechanistic optimization for a nominal influent and ten additional influent scenarios. Retained decisions are evaluated by independently solving the original mechanistic equations at the selected controls. Direct optimization gives a lower reference-model objective in all eleven retained comparisons. Across the ten influent scenarios, mean optimization times are 1.726 s for the surrogate and 5.236 s for the direct route. These times exclude model development and subsequent verification. The contribution is a connected plant surrogate that enforces specified physical relations and supports faster operating searches. The results also show that physical consistency and favorable average accuracy do not establish decision quality, which requires mechanistic evaluation.

Nomenclature

Only nonstandard mathematical symbols are listed. All vector inequalities are componentwise.

$A \in \mathbb{R}^{K \times F}$	Full-row-rank stoichiometric-invariant operator
A_{cl}	Clarifier surface area
$a_i, k_L a_i$	Normalized aeration setting and oxygen-transfer coefficient in reactor i
$c_i, m, c_E, c_U \in \mathbb{R}_+^F$	Reactor- i , mixer, clarifier-overflow, and clarifier-underflow concentrations
D_X, D_y, D_b, D_g, D_T	Positive plant-response, mechanistic-state, constraint-row, and quality scales
d_i	Recycle-adjusted dilution rate in reactor i , d^{-1}
E_S, E_X	Selectors for soluble and particulate components
F, F_S, F_X	Total, soluble, and particulate component counts
$g_E = q_E c_E, g_U = q_U c_U$	Clarifier outlet component flows divided by fresh flow
γ_S, γ_E	Ridge penalties for the raw multiresponse fit and logarithmic overflow solids model
h_i, H	Nominal stage and total hydraulic retention times on a fresh-flow basis, h
I_{comp}	Component-to-effluent-quality reporting map
J	Dimensionless engineering objective
K	Number of independent stoichiometric invariants
L, ℓ_F	Number of clarifier layers and feed-layer index
M_{cl}, V_{cl}	Clarifier solids inventory and total clarifier volume
N, n_p	Numbers of completely mixed biological stages and kinetic processes
$n_X, n_y, n_{\phi, S}, d_z$	Plant-response, mechanistic-state, feature, and statistical-input dimensions

*Corresponding author

Email address: 20102931@usc.edu.ph (E.F.S. Jr.)

ORCID(s):

q_P, q_C, q_E, q_U	Reactor-train, clarifier-feed, overflow, and underflow flow ratios
Q_0	Fresh-influent flow
r_I, r_R, w	Mixed-liquor recycle, return activated sludge, and waste activated sludge ratios relative to Q_0
s_ℓ	Mechanistic total suspended solids concentration in clarifier layer ℓ
t_X	Component-to-total-suspended-solids column vector
$X_\star$	Positive reference concentration used to nondimensionalize the logarithmic solids target
$\hat{X}_{E,\log}$	Positive overflow solids concentration fitted on the model-development set
$\vartheta \in \mathbb{R}^{d_\vartheta}$	Vector containing exactly the freely varied operating coordinates
$x \in \mathbb{R}_+^F$	Fresh-influent component vector
$x_{\text{raw},S}, \hat{x}_S$	Raw and physically projected operational responses
y	Reactor-and-clarifier mechanistic state
ν, ρ	Stoichiometric matrix and process-rate vector
ϕ_S	Standardized unique-monomial feature vector for the surrogate fit
$\hat{\beta}_{E,\gamma_E}$	Fitted coefficient vector for logarithmic overflow solids concentration
ϖ, ϖ_Q	Overall objective priorities and effluent-quality priorities

1. Introduction

Activated sludge treatment couples biological reactors with secondary clarification and sludge recirculation. Return activated sludge (RAS) carries settled biomass back to the reactors, whereas waste activated sludge (WAS) removes solids from the plant. Plants designed for biological nitrogen removal may also use mixed-liquor recycle (MLR) to return reactor liquor to upstream stages (Henze et al., 2008). Clarifier settling and thickening affect the solids returned to the reactors and the quality of the treated water (Härtel and Pöpel, 1992). Sludge age also influences oxygen-transfer efficiency, effluent particle concentrations, and removal of biodegradable organic matter (Leu et al., 2012). These coupled effects link recycle, aeration, and wasting decisions. Mathematical programming provides a framework for balancing treatment requirements, resource use, and constraints in wastewater treatment design (Padrón-Páez et al., 2020; Aboagye et al., 2021). Operational optimization applies this framework to the choice of plant settings.

Activated Sludge Models describe the biological and chemical processes through coupled nonlinear dynamic mass-balance equations (Henze et al., 2006). Settling models describe clarification and thickening (Takács et al., 1991). Time integration can approach a steady state and provide an initial state for a subsequent algebraic calculation (Kelley and Keyes, 1998; Selerio, 2026). In an activated sludge application, Selerio (2026) used dynamic relaxation to improve this initial state when the algebraic solve did not converge. When optimization evaluates candidate operating conditions through simulation, the steady-state calculation must be repeated. This repeated evaluation motivates the use of less expensive surrogate models (Bhosekar and Ierapetritou, 2018).

Statistical surrogate modeling offers an alternative to repeatedly solving the mechanistic equations during optimization. Surrogates are fitted to input–output data from experiments, mechanistic simulations, or a combination of these sources (McBride and Sundmacher, 2019). The fitted model predicts process responses at new inputs and can reduce the cost of repeated feasibility and optimization calculations (Bhosekar and Ierapetritou, 2018).

Statistical surrogates can reduce the cost of repeated process evaluation. For papermaking wastewater treatment, He et al. (2023) used a Kriging surrogate to optimize treatment performance and greenhouse gas emissions. Faster evaluation does not ensure physically meaningful predictions. Unconstrained activated sludge surrogates can predict negative component concentrations or violate stoichiometric mass conservation (Selerio, 2026).

Effluent prediction also requires attention to low residual concentrations. Longer sludge age can improve removal of biodegradable organic matter and reduce effluent particles and suspended solids (Leu et al., 2012). For a component with a low residual concentration, a small absolute prediction error can be a large relative error. For example, an error of 1 g m^{-3} is only 1% of an influent concentration of 100 g m^{-3} . The same error is 100% of a treated-water concentration of 1 g m^{-3} . Since concentrations cannot be negative, overprediction of removal can also produce a physically impossible result. The activated sludge comparison of Selerio (2026) shows that favorable aggregate prediction errors can coexist with violations of conservation and non-negativity. Prediction accuracy and physical consistency must therefore be assessed separately. Approximation quality must also be managed during the optimization search and checked against the reference model (Alexandrov et al., 1998; Bhosekar and Ierapetritou, 2018).

These risks motivate the explicit enforcement of physical constraints on statistical surrogate predictions. The purpose is to restrict predictions to states that satisfy known process requirements, rather than to assume that lower average error ensures physical consistency. Activated sludge and wastewater studies have addressed parts of this problem through hybrid models and constrained uses of surrogates. Li et al. (2024) embedded governing-equation residuals in physics-informed neural networks for unit-operation models, including an activated sludge reactor, and reported lower data requirements and better generalization across loading conditions than conventional machine learning models. At the operational level, Bernardelli et al. (2020) placed a neuro-fuzzy process model inside real-time model predictive control to select aeration while jointly pursuing nutrient-removal and energy objectives. At the broader wastewater-process level, Durkin et al. (2024) used data-driven unit surrogates within constrained superstructure optimization for resource recovery.

These approaches differ in which constraints they impose and where they impose them. A residual penalty can reduce violations, but exact conservation at a feasible prediction requires conservation to be imposed as an equality. Analytic output constraints provide one way to impose such relations by construction (Beucler et al., 2021). Bounds on operating decisions likewise do not impose a conservation relation on surrogate outputs unless that relation is explicitly enforced. In activated sludge modeling, Selerio (2026) projected second-order regression predictions onto mass-conservation and component-nonnegativity constraints. This approach retains a statistical input–output model while ensuring that each projected prediction satisfies the enforced constraints to numerical tolerance.

Related studies outside activated sludge modeling show that physical constraints can be introduced at different stages of surrogate construction. Analytic conservation constraints have been incorporated into neural network output mappings for atmospheric convection (Beucler et al., 2021). For atmospheric photochemistry, Sturm and Wexler (2020) formulated conserved fluxes as learning targets. Sturm and Wexler (2022) later embedded the stoichiometric mapping as a fixed layer that conserves atoms to numerical precision. Other studies have used constrained training data and a mass-balance penalty to reduce conservation errors (Ruckstuhl et al., 2021). Physical consistency can also be imposed after an unconstrained model has made a prediction. Sturm and Silva (2025) applied a species-weighted correction that enforces atom conservation. Valente et al. (2025) projected surrogate outputs onto manifolds defined by conservation laws. These applications concern atmospheric and plasma systems. The activated sludge projection of Selerio (2026) uses the same principle of correcting predictions to satisfy known balances.

Differentiable projection has also been incorporated into neural surrogate models. ENFORCE combines learned predictions with adaptive projection for nonlinear equality and inequality constraints; its nonlinear convergence analysis is conditional on local regularity, whereas affine constraints have stronger projection properties (Lastrucci and Schweidtmann, 2026). Physics-consistent neural operators apply projections to spatial fields, including Fourier-space mass and momentum constraints for spatiotemporal forecasting (Xu et al., 2025). For a steady-state treatment plant, the projected response must instead connect unit concentrations, recycle streams, component-resolved separation, and stored solids. The formulation considered here represents these quantities jointly and reduces clarifier inventory under a specified layer envelope. Applying the same plant projection to different statistical predictors allows their approximation errors to be compared under common physical constraints.

Within the studies reviewed here, two relevant capabilities remain separate. The Invariant-Constrained Second-Order Regression (ICSOR) formulation of Selerio (2026) projects the response of one continuous stirred-tank reactor (CSTR) onto conservation and non-negativity constraints. It does not represent the mixer, reactor train, secondary clarifier, and recycle streams as one coupled response. He et al. (2023) demonstrates surrogate-assisted optimization of a wastewater treatment process but does not impose hard plant-wide projection of connected component responses. This study combines projection of plant-wide outputs with surrogate-based operating optimization. Physical consistency within one CSTR does not establish consistency of recycle mixing or clarifier outlet balances across the plant. The proposed extension must therefore preserve the physical connections among units and verify selected decisions against a mechanistic reference. Physical feasibility of a surrogate prediction must remain distinct from prediction accuracy and decision quality.

This study develops and evaluates a plant-wide, physically constrained statistical surrogate based on ICSOR for steady-state operating optimization. The contribution combines a joint response that preserves mixer–reactor–clarifier connectivity, a tight reduction of clarifier inventory conditional on the adopted layer envelope, and independent mechanistic evaluation of selected decisions. Three questions guide the evaluation. The first examines how projection affects physical consistency and prediction accuracy across the plant. The second asks whether surrogate-selected controls retain their predicted performance under the mechanistic model. The third compares decision quality and computational cost with direct mechanistic optimization under common operating choices and priorities. Constraint

ablations, predictor comparisons, and diagnostic sensitivity analyses distinguish the contributions of the statistical approximation, imposed physical relations, and empirical screening. Initialization and cost analyses examine the computational implications of repeated projection and surrogate-assisted mechanistic optimization. Because the analysis choices were informed by inspection of the simulation data, the evaluation provides descriptive post-selection evidence (Kaufman et al., 2012). Its scope is the specified plant topology, operating domain, and objective family.

2. Theory and calculation

The modeling problem is to select reactor capacity, aeration, recycle, and wasting settings for a coupled activated sludge plant at steady state. Each choice affects both local unit conditions and the material returned through the recycle streams. The formulation therefore begins with hydraulic and component balances that connect the mixer, reactors, and clarifier. These relations define the physical requirements used to project the statistical predictions onto a physically feasible set. The resulting plant response supplies the treatment and resource measures used in the objective and engineering limits. The section then develops the surrogate optimization problem and a direct mechanistic comparator. Numerical solution, route-specific normalization, and retrospective verification are described in the methodology. Both routes are judged on the same mechanistic reference without assuming that projection reproduces the full model.

2.1. Plant configuration and hydraulic balances

The hydraulic balances establish how each recycle setting changes the flow through the treatment units. Consider $N \geq 1$ completely mixed biological stages followed by an $L \geq 3$ layer clarifier. The component representation distinguishes soluble transport from particulate settling and thickening (Takács et al., 1991; Jeppsson and Diehl, 1996). Component vectors contain $F = F_S + F_X$ entries. The matrices $E_S \in \mathbb{R}^{F_S \times F}$ and $E_X \in \mathbb{R}^{F_X \times F}$ select the disjoint soluble and particulate entries, so $E_S^\top E_S + E_X^\top E_X = I_F$ and $E_S E_X^\top = 0$. The non-negative total suspended solids (TSS) map t_X has zero soluble entries. Figure 1 shows the plant connections. The final reactor supplies both the MLR stream and the clarifier. Clarifier underflow is split into RAS, which returns internally, and WAS, which leaves the plant.

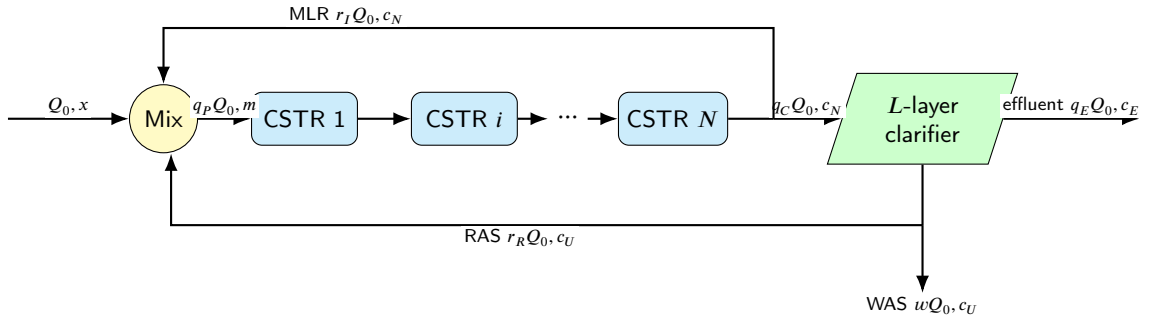


Figure 1: General recycling mixer-reactor-clarifier system. Internal recycle streams change unit conditions but cancel from the external plant balance.

Normalizing every flow by the fresh-influent flow Q_0 gives

$$q_P = 1 + r_I + r_R, \quad q_C = 1 + r_R, \quad q_U = r_R + w, \quad q_E = 1 - w. \quad (1)$$

The clarifier identity $q_C = q_E + q_U$ follows immediately. The operating domain requires $r_I \geq 0$, $r_R \geq 0$, $0 \leq w < 1$, and $q_U = r_R + w > 0$, so both outlet concentrations can later be recovered safely from their mass flows. At the mixer, fresh feed, MLR, and RAS meet. Dividing its component balance by Q_0 gives

$$q_P m = x + r_I c_N + r_R c_U. \quad (2)$$

Recycle changes the flow through a reactor without changing its volume. The nominal hydraulic retention time (HRT) based on fresh flow must therefore be distinguished from the actual stage residence time. Define

$$h_i = \frac{24V_i}{Q_0}, \quad H = \sum_{i=1}^N h_i, \quad d_i = \frac{Q_0 q_P}{V_i} = \frac{24q_P}{h_i}, \quad (3)$$

Here h_i and H are in hours, and d_i is in d^{-1} . The actual stage residence time is h_i/q_p hours. Increasing recycle at fixed volume therefore increases dilution and reduces the time spent in each pass through a reactor. Individual stage volumes may be varied through h_i , or fixed volume fractions may be imposed through $h_i = f_i^V H$ with $\sum_i f_i^V = 1$. Equal tanks correspond to $f_i^V = 1/N$.

The optimization variables must correspond to the choices available in the study. Let $\vartheta \in \mathbb{R}^{d_\vartheta}$ contain only these freely varied quantities. They may include selected h_i , aeration settings a_i , stage-addition rates, and the three recycle and wasting ratios. A stage addition represented only by a source term is assumed to have negligible carrier flow. If its liquid flow is not negligible relative to Q_0 , that stream must also enter the hydraulic balances and change the downstream flow and dilution rates. Fixed settings remain parameters rather than zero-variance regression inputs. This definition avoids fitting coefficients for settings that never vary.

2.2. Reactor balances and conserved inventories

The hydraulic relations determine the dilution rate used in each reactor balance. These balances connect the operating settings to biological conversion and provide conservation relations for the surrogate. In reactor i , material enters from the preceding stage, leaves at the train flow, and is transformed by biological and chemical processes (Henze et al., 2006). Oxygen transfer and declared external additions provide separate sources. With $c_0 = m$, the steady component balance is

$$0 = d_i(c_{i-1} - c_i) + v^\top \rho_i(c_i; \vartheta) + \omega_i(a_i, c_i)e_{S_O} + \xi_i(\vartheta), \quad i = 1, \dots, N. \quad (4)$$

The matrix $v \in \mathbb{R}^{n_p \times F}$ stores the stoichiometric changes caused by n_p processes, and $\rho_i \in \mathbb{R}_+^{n_p}$ contains their volumetric rates. The vector e_{S_O} selects dissolved oxygen (DO), and $\omega_i = k_L a_i(a_i)(S_O^* - c_{i,S_O})$ describes oxygen transfer. The term ξ_i represents an explicitly modeled stage addition with negligible carrier flow. Every term has units of component concentration per day. Retaining ξ_i ensures that an optimized carbon or chemical dose enters the material balance. At a zero component concentration, the combined reaction and source terms are required to be non-negative. For a locally Lipschitz dynamic balance with non-negative initial conditions and inflows, this boundary condition preserves non-negative concentrations (Haddad et al., 2010). This property of the continuous model does not guarantee non-negative numerical integration steps.

The surrogate should preserve conserved inventories without solving every reaction-rate equation during prediction. Some linear combinations of components are changed neither by the reaction network nor by oxygen transfer (Sturm and Wexler, 2022; Kircher and Votsmeier, 2025). Let $A \in \mathbb{R}^{K \times F}$ contain independent, physically identified inventory rows satisfying

$$Av^\top = 0, \quad Ae_{S_O} = 0, \quad \text{rank}(A) = K, \quad \|A_{k,:}\|_2 = 1. \quad (5)$$

Premultiplication of Equation (4) by A removes the reaction and aeration terms and gives the stagewise inventory balance

$$A(c_i - c_{i-1}) = d_i^{-1} A \xi_i(\vartheta), \quad i = 1, \dots, N. \quad (6)$$

When $\xi_i = 0$, the quantity Ac_i is constant across the stage. Reactions can redistribute a conserved inventory among components but cannot create or remove it. These relations prevent the surrogate from improving a predicted treatment response by violating an inventory balance. If a stage dose adds a conserved quantity, Equation (6) accounts for that addition. Its right-hand side can remain zero only when $A \xi_i = 0$.

Figure 2 distinguishes the geometric roles of the mixer and reactor balances. The mixer concentration is a convex combination of its incoming compositions. In a reactor without external additions, conversion preserves the affine set defined by $Ac_i = Ac_{i-1}$. Reaction directions are tangent to this set, while the invariant rows are normal to it. The kinetic equations determine which points on the set represent steady operation.

2.3. Clarifier outlet balances and aggregate inventory

The final reactor supplies the clarifier feed. The clarifier outlet balances complete the recycle connection to the mixer and determine the component flows leaving the plant. Since q_U may be small, these balances are written in outlet component flows rather than outlet concentrations. Define

$$g_E = q_E c_E, \quad g_U = q_U c_U. \quad (7)$$

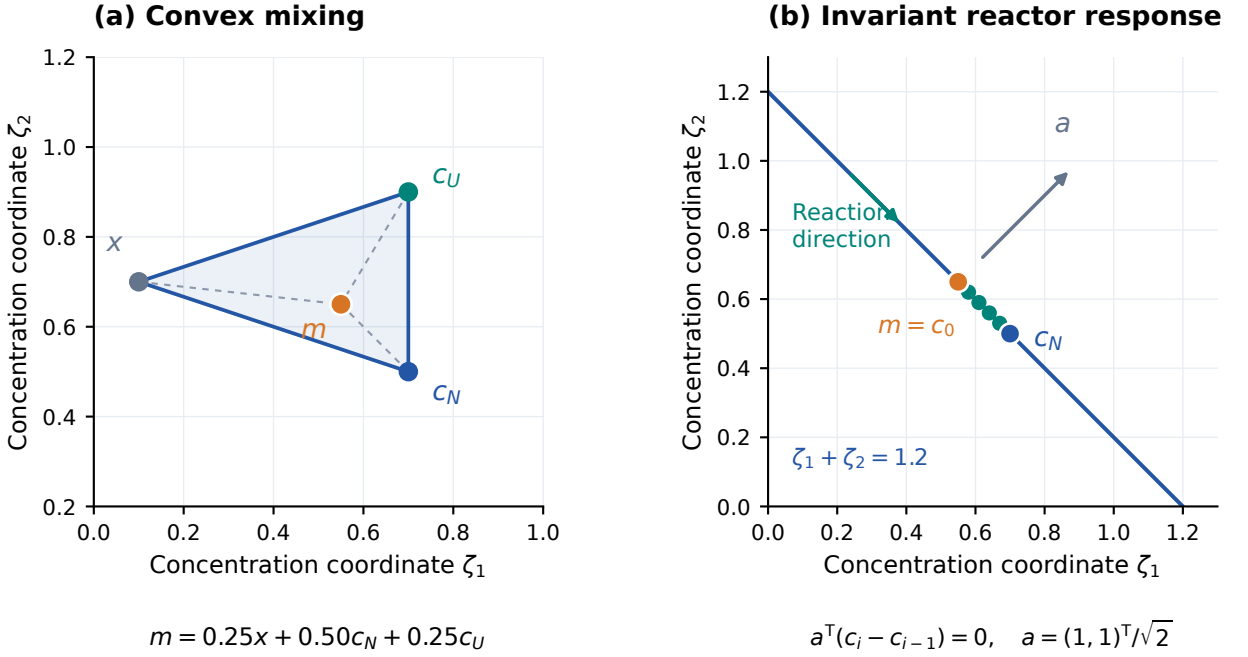


Figure 2: Dimensionless two-component illustration of network balances. (a) For $r_I = 2$ and $r_R = 1$, Equation (2) gives $m = 0.25x + 0.50c_N + 0.25c_U$, inside the convex hull of the incoming compositions. (b) With no external addition and invariant normal $a = (1, 1)^T/\sqrt{2}$, the illustrated reactor responses remain on $\zeta_1 + \zeta_2 = 1.2$. The reaction direction is orthogonal to a . The points illustrate conservation geometry rather than solutions of the reaction-rate equations.

The component balance around the complete vessel is then

$$q_C c_N = g_E + g_U. \quad (8)$$

Because $q_E, q_U > 0$, the concentrations are recovered as $c_E = g_E/q_E$ and $c_U = g_U/q_U$. Using flows in the balance avoids dividing by a small underflow ratio within the conservation constraint.

The clarifier is treated here as a nonreactive separator, following the settling-model approach of Takács et al. (1991). Under this assumption, soluble material follows the liquid and particulate material can concentrate in the underflow. The linear relations retained in the projection are

$$E_S g_U = q_U E_S c_N, \quad E_X g_U \geq q_U E_X c_N. \quad (9)$$

The first relation, together with Equation (8), also gives $E_S g_E = q_E E_S c_N$. Thus the clarifier does not remove soluble material through settling. The second relation prevents an underflow particulate concentration below the feed concentration. With non-negative outlet flows, it bounds the underflow recovery of each particulate component with positive feed concentration between q_U/q_C and one. Applying the non-negative TSS map and the total clarifier balance then gives $X_E \leq X_N \leq X_U$, where $X_N = t_X^T c_N$ is the feed TSS concentration.

Outlet concentrations determine the returned and discharged solids, but they do not determine the solids stored inside the clarifier. This stored mass is needed to calculate whole-plant solids age. Let s_ℓ be the mechanistic TSS concentration in layer ℓ , numbered from the water surface ($\ell = 1$) to the hopper ($\ell = L$). The endpoint concentrations follow from the outlet component flows as

$$X_E = s_1 = \frac{t_X^T g_E}{q_E}, \quad X_U = s_L = \frac{t_X^T g_U}{q_U}. \quad (10)$$

The mechanistic acceptance envelope

$$X_E \leq s_\ell \leq X_U, \quad \ell = 2, \dots, L-1, \quad (11)$$

keeps every internal layer between the two outlet concentrations. It is intentionally an endpoint envelope, not a claim of layer-by-layer monotonicity. The nonlinear settling-flux balances, which determine the detailed shape, remain part of the mechanistic model. With positive layer volumes, the layer values determine the physical inventory

$$M_{cl} = \sum_{\ell=1}^L V_{cl,\ell} s_{\ell}, \quad V_{cl} = \sum_{\ell=1}^L V_{cl,\ell}. \quad (12)$$

Only this scalar inventory is retained in the statistical response. Eliminating the unneeded internal layer coordinates from Equations (10)–(12) gives the tight interval

$$\begin{aligned} \underline{M}_{cl} &= (V_{cl} - V_{cl,L})X_E + V_{cl,L}X_U, \\ \overline{M}_{cl} &= V_{cl,1}X_E + (V_{cl} - V_{cl,1})X_U, \\ \underline{M}_{cl} &\leq M_{cl} \leq \overline{M}_{cl}. \end{aligned} \quad (13)$$

For the ordered endpoints $X_E \leq X_U$, the interval is necessary and sufficient for the existence of at least one profile satisfying the endpoint envelope and having inventory M_{cl} . The lower bound is attained when every internal layer is assigned X_E , and the upper bound is attained when every internal layer is assigned X_U . Assigning all internal layers a common value between these endpoints changes the inventory continuously between the two bounds. These bounds prevent the predicted stored mass from contradicting the predicted outlet concentrations. They retain useful storage information without requiring a statistical prediction for every layer. An admissible inventory does not establish that any corresponding profile satisfies the nonlinear settling-flux equations. That check remains with the mechanistic model.

The geometric reduction in Figure 3 maps many admissible layer profiles to one inventory interval. The extremal profiles attain its endpoints, and profiles inside the envelope can have nonmonotone layer concentrations. The width of the interval decreases to zero as the two outlet concentrations approach one another.

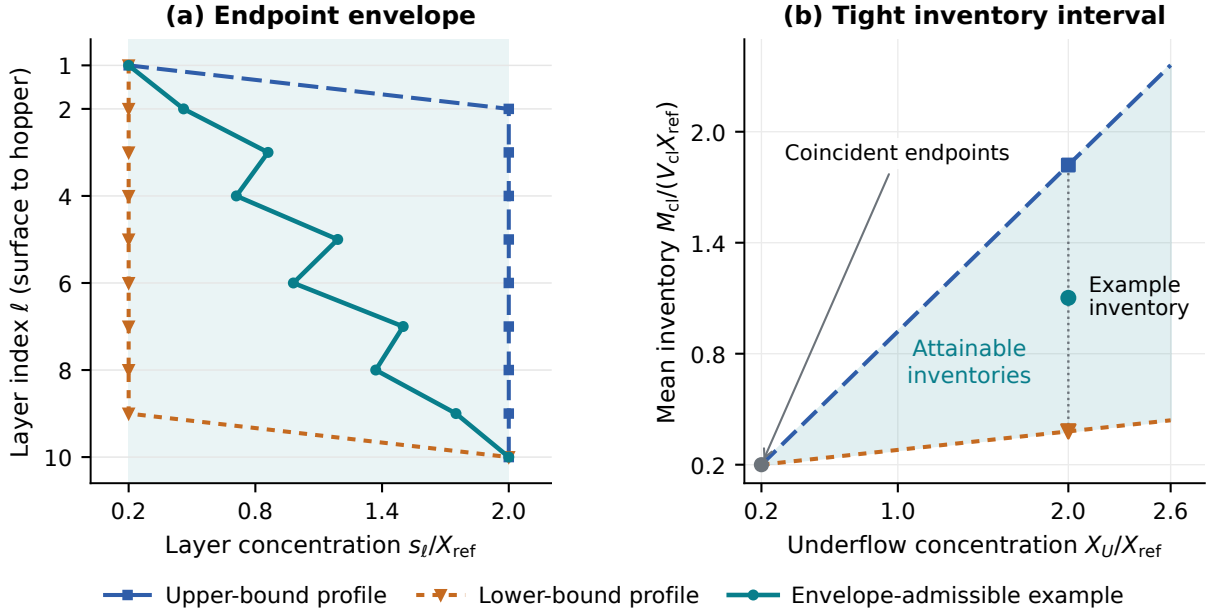


Figure 3: Clarifier inventory geometry for ten equal-volume layers and an arbitrary positive concentration scale X_{ref} . (a) At $X_E/X_{ref} = 0.2$ and $X_U/X_{ref} = 2$, a nonmonotone example satisfies the endpoint envelope. Assigning every internal layer to either endpoint gives the lower and upper inventory extremizers; connecting segments only guide the eye between discrete layers. (b) At fixed $X_E/X_{ref} = 0.2$, the shaded band contains exactly the normalized mean inventories allowed by Equation (13). The two bounds coincide when $X_U = X_E$. Envelope-admissible profiles need not satisfy the nonlinear settling-flux equations.

2.4. Raw plant-response regression and overflow solids model

The preceding balances identify the quantities needed to represent the connected plant without predicting every clarifier layer. The extended ICSOR surrogate (Selerio, 2026), designated route S , therefore predicts the mixer and reactor states, outlet component flows, and aggregate inventory together. These quantities supply the objective and constraint calculations developed below. The response vector is ordered in process sequence as

$$\chi = \text{col}(m, c_1, \dots, c_N, g_E, g_U, M_{cl}) \in \mathbb{R}^{n_\chi}, \quad n_\chi = (N + 3)F + 1. \quad (14)$$

The endpoint TSS values remain recoverable from g_E and g_U , while M_{cl} preserves the storage information required for solids retention time (SRT). The full L -layer profile remains in the mechanistic target from which this reduced response is constructed.

The regression includes interactions among influent composition and operating settings. For example, oxygen demand depends on the load entering the reactors (Henze et al., 2008). Centering and scaling put the regression features on comparable scales before ridge regularization (Hastie et al., 2009). Let $z = \text{col}[D_{\theta, \text{fit}}^{-1}(\vartheta - \bar{\vartheta}), D_{x, \text{fit}}^{-1}(x - \bar{x})] \in \mathbb{R}^{d_z}$, where $d_z = d_\vartheta + F$ and the means and positive diagonal scales are computed from the model-development data. To include curvature and pairwise interactions without duplicate terms, form

$$r(z) = \text{col}[z, \text{vech}(zz^T)], \quad \phi_S(\vartheta, x) = \text{col}[1, D_{r, \text{fit}}^{-1}\{r(z) - \bar{r}\}] \in \mathbb{R}^{n_{\phi, S}}, \quad (15)$$

where vech lists one triangular copy of the symmetric quadratic matrix. Hence

$$n_{\phi, S} = 1 + d_z + \frac{d_z(d_z + 1)}{2}. \quad (16)$$

With response mean $\bar{\chi}$ and positive diagonal scale matrix D_χ computed from the model-development set, the route- S raw response is

$$\chi_{\text{raw},S}(\vartheta, x) = \bar{\chi} + D_\chi \hat{C}_{S,\gamma_S} \phi_S(\vartheta, x). \quad (17)$$

The ridge parameter γ_S controls a squared-coefficient penalty that stabilizes estimation when features are correlated (Hastie et al., 2009). Equation (17) is still an unconstrained regression prediction. It does not enforce sign or balance requirements (Kircher and Votsmeier, 2025).

Overflow TSS spans a strongly asymmetric range. Small absolute errors at low treated-water concentrations can correspond to large relative errors. A fit over the full plant response may give these errors little weight when other responses have much larger magnitudes. A separate logarithmic model therefore places the endpoint on a relative-error scale without changing the raw multiresponse fit (Tofallis, 2015). Define the dimensionless logarithmic training target

$$\ell_E = \log\left(\frac{X_E}{X_\star}\right), \quad X_\star = 1 \text{ mg L}^{-1}, \quad (18)$$

and fit a separate scalar ridge coefficient vector $\hat{\beta}_{E,\gamma_E}$ on the same complete second-order input family. Its positive concentration-scale prediction is

$$\hat{X}_{E,\log}(\vartheta, x) = X_\star \exp\left\{\hat{\beta}_{E,\gamma_E}^\top \phi_S(\vartheta, x)\right\} > 0. \quad (19)$$

The logarithm assigns the same discrepancy to a twofold error at low or high concentration. Exponentiation prevents a negative fitted endpoint. The positive prediction varies with all declared controls and influent coordinates and is not a mechanistic lower bound. The fitting procedure is described in the methodology and the Supplementary Material, Section S3.

2.5. Joint projection of the plant response

The two regressions supply a raw plant response and a positive overflow-TSS estimate, but neither enforces the network balances. A joint projection adjusts these predictions to satisfy the physical relations derived above (Kircher and Votsmeier, 2025; Valente et al., 2025). Correcting the complete response together preserves the connections between units and recycle streams. The physical balances remain distinct from the empirical overflow closure. Define $\delta_i(\vartheta) = d_i^{-1} A \xi_i(\vartheta)$ and let $\mathcal{H}(\vartheta)\chi = \mathbf{b}(\vartheta, x)$ represent

$$\text{col}\left(\begin{array}{c} q_P m - r_I c_N - (r_R/q_U)g_U - x \\ \{A(c_i - c_{i-1}) - \delta_i\}_{i=1}^N, \quad c_0 = m \\ g_E + g_U - q_C c_N \\ E_S g_U - q_U E_S c_N \end{array}\right) = 0. \quad (20)$$

The network balances alone do not determine the degree of solids separation. The fitted overflow endpoint supplies this additional empirical relation through

$$t_X^\top g_E = q_E \hat{X}_{E,\log}(\vartheta, x). \quad (21)$$

At fixed (ϑ, x) , the right-hand side is a known positive scalar, so Equation (21) is linear in χ and preserves the quadratic program (QP) structure. The complete equality operator $\mathcal{H}_{\log}(\vartheta)\chi = \mathbf{b}_{\log}(\vartheta, x)$ collects Equations (20) and (21). There are

$$n_h = 2F + NK + F_S + 1 \quad (22)$$

rows covering the mixer balance, N invariant transitions, the full clarifier balance, soluble pass-through, and the scalar overflow-TSS closure. The closure fixes only total overflow TSS. It does not prescribe the particulate composition of g_E . That allocation is determined by joint projection of the raw component predictions subject to the remaining constraints.

Particulate densification and inventory bounds restrict responses that could satisfy the equalities but still contradict the adopted settling envelope. These inequalities are written as

$$\mathcal{G}(\vartheta)\chi = \text{col}\left(\begin{array}{c} q_U E_X c_N - E_X g_U \\ q_U(V_{\text{cl}} - V_{\text{cl},L})t_X^\top g_E + q_E V_{\text{cl},L}t_X^\top g_U - q_E q_U M_{\text{cl}} \\ q_E q_U M_{\text{cl}} - q_U V_{\text{cl},L}t_X^\top g_E - q_E(V_{\text{cl}} - V_{\text{cl},L})t_X^\top g_U \end{array}\right) \leq 0. \quad (23)$$

Thus $n_g = F_\chi + 2$ inequality rows describe particulate densification and aggregate-inventory feasibility. The last two rows are the division-free form of Equation (13) and remain linear in χ at fixed ϑ . Non-negativity contributes another n_χ inequalities.

The correction is measured after scaling each response by D_χ^{-1} , so that the large-magnitude inventory coordinate does not overwhelm a small nutrient coordinate (Sturm and Silva, 2025). The projected displacement and response are (Selerio, 2026; Valente et al., 2025)

$$\hat{u}_S(\vartheta, x) = \arg \min_{u \in \mathbb{R}^{n_\chi}} \frac{1}{2} \|u\|_2^2 \quad (24)$$

$$\begin{aligned} \text{subject to } & D_b^{-1} \{H_{\log}(\vartheta)[\chi_{\text{raw},S} + D_\chi u] - \mathbf{b}_{\log}(\vartheta, x)\} = 0, \\ & \chi_{\text{raw},S} + D_\chi u \geq 0, \\ & D_g^{-1} \mathcal{G}(\vartheta)[\chi_{\text{raw},S} + D_\chi u] \leq 0, \\ \hat{\chi}_S(\vartheta, x) &= \chi_{\text{raw},S}(\vartheta, x) + D_\chi \hat{u}_S(\vartheta, x). \end{aligned} \quad (25)$$

D_b and D_g rescale the constraint rows without changing the feasible set. This scaling improves numerical conditioning (Boyd and Vandenberghe, 2004, Section 11.8.2). The overflow-closure row and the other equality rows receive positive scales computed from the model-development set. At fixed inputs, the linear constraints define a closed convex set. If this set is nonempty, the strictly convex, coercive objective has a unique minimizer (Boyd and Vandenberghe, 2004). Because Equation (21) is fitted rather than a physical identity, feasibility is not guaranteed throughout the operating box. An accepted projection supplies an audited feasible witness. A rejected QP solve or a failed optimality audit alone does not establish that the constraint set is empty: iteration limits, conditioning, or multiplier reconstruction can also cause rejection. The numerical assessment distinguishes audited success, infeasibility diagnosed by an independent linear-feasibility solve, and unresolved numerical failure. Attribution to the empirical closure additionally requires an audited feasible witness after removing only Equation (21) at the same input. Solver diagnoses remain subject to numerical tolerances. The operating route rejects unaudited responses, and the hard closure is not clipped or relaxed.

Enforcing constraints does not by itself establish greater prediction accuracy (Sturm and Silva, 2025). Euclidean projection onto a nonempty closed convex set cannot increase the distance to a reference in that set (Boyd and Vandenberghe, 2004). Here, the distance is measured in the standardized response coordinates used in Equation (24). However, the mechanistic overflow TSS generally differs from the fitted value in Equation (21), so that guarantee is not presumed here. The error audit therefore accounts for the reference's distance to the set, as detailed in the Supplementary Material, Section S4.2. The projection does not reconstruct a clarifier profile or enforce all kinetic and nonlinear settling equations. Those equations are checked separately by evaluation with the complete mechanistic model.

In coordinates $\zeta = D_\chi^{-1} \chi$, Equation (24) minimizes the Euclidean distance from the standardized raw response to the feasible set. Figure 4 shows this nearest-point geometry and the effect of an additional closure equality. Positivity of a closure target alone does not ensure that its equality intersects the remaining constraints.

2.6. Constraint ablations and alternative predictors

At a fixed input, the mixer and invariant balances and the nonreactive clarifier identities are affine relations derived from the declared process assumptions. They are not linearizations of the complete nonlinear kinetics or settling fluxes. The densification and inventory restrictions add separation assumptions, and the fitted overflow equality adds an empirical closure. These distinct roles motivate controlled removal of constraint groups. The full model retains every group. Removing densification deletes the componentwise particulate inequalities; removing inventory deletes the two storage bounds; removing the closure deletes only its equality. A basic conservation projection retains mixer, stage-invariant, and clarifier component balances and non-negativity while omitting soluble pass-through and the three additional groups. Response coordinates, raw predictions, and the projection metric are held fixed so that a comparison concerns the retained constraints.

The inventory interval implies aggregate ordering of the TSS endpoints when the total volume of internal layers is positive. Removing componentwise densification while retaining this interval therefore preserves some separation information. Whole-plant SRT is descriptive in the present operating problem. Inventory bounds directly affect stored-solids and SRT fidelity, while their influence on selected controls acts through the joint projection and trust diagnostics. Since projection adjusts all response coordinates together, a constraint can reduce error in one coordinate while increasing it in another.

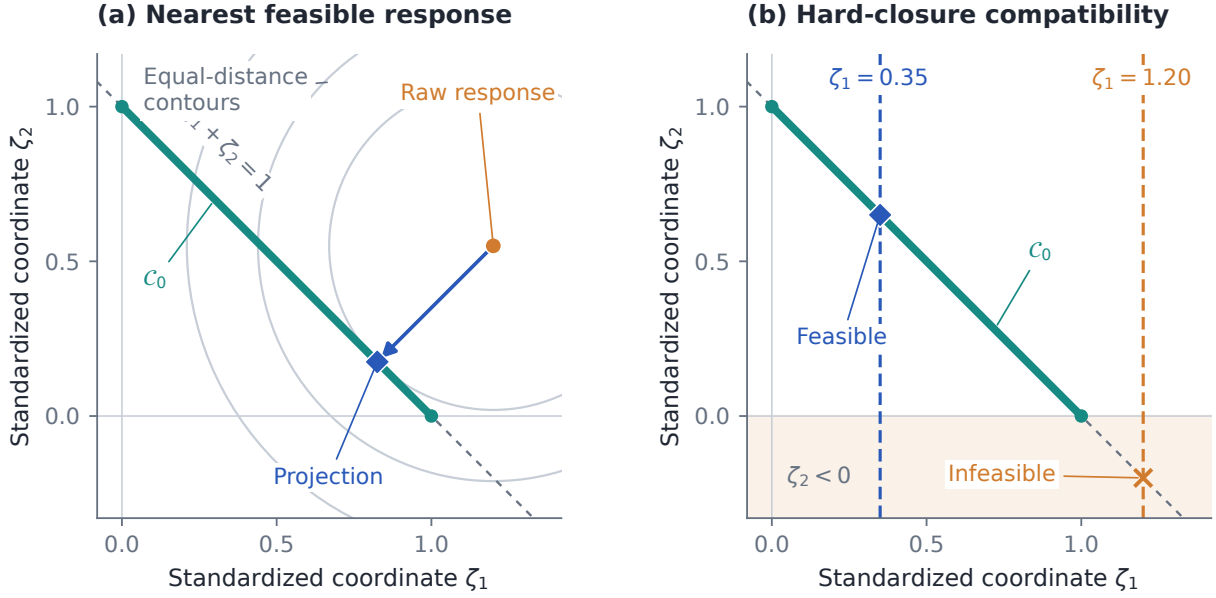


Figure 4: Projection and closure compatibility in an illustrative standardized response space. The constraint set is $C_0 = \{\zeta \in \mathbb{R}_+^2 : \zeta_1 + \zeta_2 = 1\}$. (a) The raw response $\zeta_{\text{raw}} = (1.20, 0.55)$ projects to $\zeta^* = (0.825, 0.175)$; circles denote equal distance from ζ_{raw} . The dashed line extends the affine balance beyond its nonnegative segment. (b) Adding $\zeta_1 = 0.35$ gives the feasible point $(0.35, 0.65)$. Adding $\zeta_1 = 1.20$ instead requires $\zeta_2 = -0.20$, so the intersection is empty despite a positive closure target.

A Gaussian-kernel ridge predictor with a Nyström approximation provides an alternative to quadratic regression (Williams and Seeger, 2001). It predicts the same reduced response from the same controls and influent variables. Cross-validation on the development set determines its bandwidth and regularization. Raw predictions and predictions corrected by the full plant projection are assessed separately. Holding the closure, physical constraints, response metric, and evaluation inputs fixed isolates the effect of the predictor family. The comparison considers prediction error, physical consistency, and projection acceptance. The kernel formulation and fitting settings are given in the Supplementary Material, Section S9.

2.7. Engineering objective and operating constraints

The projected response provides the concentrations, outlet flows, and stored solids needed to evaluate operating decisions. Effluent-quality targets influence reactor sizing, oxygen supply, recycle, and solids management (Henze et al., 2008). The objective combines treatment and resource measures to represent these competing requirements (Araujo et al., 2013; Padrón-Páez et al., 2020). Their relative priorities are case-study choices rather than universal constants. Let $z_E = I_{\text{comp}} c_E \in \mathbb{R}_+^4$ contain effluent chemical oxygen demand (COD), total nitrogen (TN), total phosphorus (TP), and TSS. With positive normalization D_T and quality priorities $\varpi_Q \geq 0$, $\mathbf{1}^\top \varpi_Q = 1$, define

$$j_Q = \varpi_Q^\top D_T^{-1} z_E. \quad (26)$$

Representative dimensionless resource terms are

$$\begin{aligned} j_H &= \frac{H - H^L}{H^U - H^L}, & j_A &= \frac{\sum_{i \in A} V_i k_L a_i}{E_A^{\text{ref}}}, \\ j_I &= \frac{r_I - r_I^L}{r_I^U - r_I^L}, & j_R &= \frac{r_R - r_R^L}{r_R^U - r_R^L}, & j_W &= \frac{w t_X^\top c_U}{w^U X_U^{\text{ref}}}. \end{aligned} \quad (27)$$

$\mathcal{A}$ is the set of stages whose aeration can be nonzero. The fixed normalization $E_A^{\text{ref}} > 0$ is chosen before optimization. In the case study, it is the maximum value of $\sum_{i \in \mathcal{A}} V_i k_L a_i$ over the declared operating box. The aeration term therefore measures oxygen-transfer capacity at the selected volumes and aeration settings. It does not represent the actual oxygen-transfer rate, which also depends on the DO deficit. Its denominator does not change when stage volumes are varied. This prevents an increase in capacity from being offset merely by rescaling the objective. These resource terms are engineering proxies rather than a monetary cost model. Additional declared operating terms can be included in the same manner. With $j = \text{col}(j_Q, j_H, j_A, j_I, j_R, j_W)$, the objective is

$$J(\vartheta, \chi; \varpi) = \varpi^\top j(\vartheta, \chi), \quad \varpi \geq 0, \quad \mathbf{1}^\top \varpi = 1. \quad (28)$$

The vectors ϖ and ϖ_Q distinguish overall operating priorities from priorities among effluent-quality measures. Their values are stated and justified in the case study. During optimization, D_T denotes the fixed normalization of the selected route. Both reference-model objectives instead use the direct-route normalization. The case-study definition states the implemented scales explicitly.

An objective alone could favor settings that exceed the plant's operating limits (Araujo et al., 2013). Engineering constraints therefore restrict solids storage, clarifier loading, and outlet conditions using the same response as the objective. Define $X_i = t_X^\top c_i$ and use X_E and X_U from Equation (10). The external solids loss rate is

$$\dot{M}_X = Q_0(q_E X_E + w X_U). \quad (29)$$

The whole-plant SRT, surface overflow rate (SOR), and clarifier solids loading rate (SLR) are then

$$\text{SRT} = \frac{\sum_{i=1}^N V_i X_i + M_{\text{cl}}}{\dot{M}_X}, \quad \dot{M}_X \geq \epsilon_X > 0, \quad (30)$$

$$\text{SOR} = \frac{Q_0 q_E}{A_{\text{cl}}}, \quad (31)$$

$$\text{SLR} = \frac{10^{-3} Q_0 q_C X_N}{A_{\text{cl}}}. \quad (32)$$

The factor 10^{-3} converts grams to kilograms in SLR. Including M_{cl} in Equation (30) accounts for solids stored outside the biological reactors. Omitting this inventory would underestimate whole-plant SRT for the same solids loss rate. The case-specific engineering constraints form $g_{\text{eng}}(\vartheta, \chi) \leq 0$. They restrict underflow TSS, prevent a nearly solids-free clarifier feed, and require positive external solids loss. Their definitions and values are provided in the Supplementary Material, Section S5. Whole-plant SRT, SOR, and SLR remain descriptive operating measures.

Network feasibility does not prevent an optimizer from selecting a response that needs an unusually large correction or is inconsistent with reaction kinetics. The additional constraints $g_{\text{trust},S}$ restrict such behavior using four diagnostics calibrated on the model-development set. The standardized correction $\kappa_{\text{corr},S} = \|\hat{u}_S\|_2 / \sqrt{n_\chi}$ measures how far projection moves the raw response. Feature leverage is commonly used to determine whether a prediction lies within the fitted input domain (Sahigara et al., 2012). The regularized form used here measures support from the fitted input features. Deviation from a common particulate recovery tests consistency with the mechanistic outlet-composition assumption. The scaled smooth reactor-balance residual measures disagreement with reactor kinetics. These diagnostics provide screening rather than a guarantee of mechanistic accuracy. Their formulas and calibration rules are given in the Supplementary Material, Section S4.3. Correction, recovery, and reactor limits use out-of-fold 95th percentiles. The leverage limit uses the maximum value among model-development rows in the final fit. Neither calculation uses holdout or optimization responses. Nonlinear clarifier behavior is checked only through evaluation with the original L -layer model because the aggregate response cannot supply layer-flux residuals.

The extended-ICSOR operational problem combines these requirements as

$$\begin{aligned} \min_{\vartheta} \quad & J[\vartheta, \hat{\chi}_S(\vartheta, x); \varpi] \\ \text{subject to} \quad & \vartheta \in \mathcal{V}, \\ & g_{\text{eng}}[\vartheta, \hat{\chi}_S(\vartheta, x)] \leq 0, \\ & g_{\text{trust},S}[\vartheta, \chi_{\text{raw},S}, \hat{\chi}_S] \leq 0, \\ & \hat{\chi}_S(\vartheta, x) \text{ solves Equation (24).} \end{aligned} \quad (33)$$

Here $\mathcal{V}$ contains the declared bounds and any stage-allocation equalities. The projected response is unique whenever the lower problem is feasible. The upper problem is generally nonconvex and piecewise smooth because the projection QP can change its active set (Tøndel et al., 2003; Colson et al., 2007). Accordingly, ϑ_S denotes the best feasible candidate retained by the search, together with its reported convergence tier. It is not a certified global minimizer.

Figure 5 illustrates how a projected response can remain continuous while its derivative changes at an active-set transition. For a fixed polyhedron and an affine raw-response path, the projection is piecewise affine. In the plant model, the constraint operators and right-hand side also depend on the controls; local differentiability therefore requires regularity of the active constraints and verification of the derivative calculation.

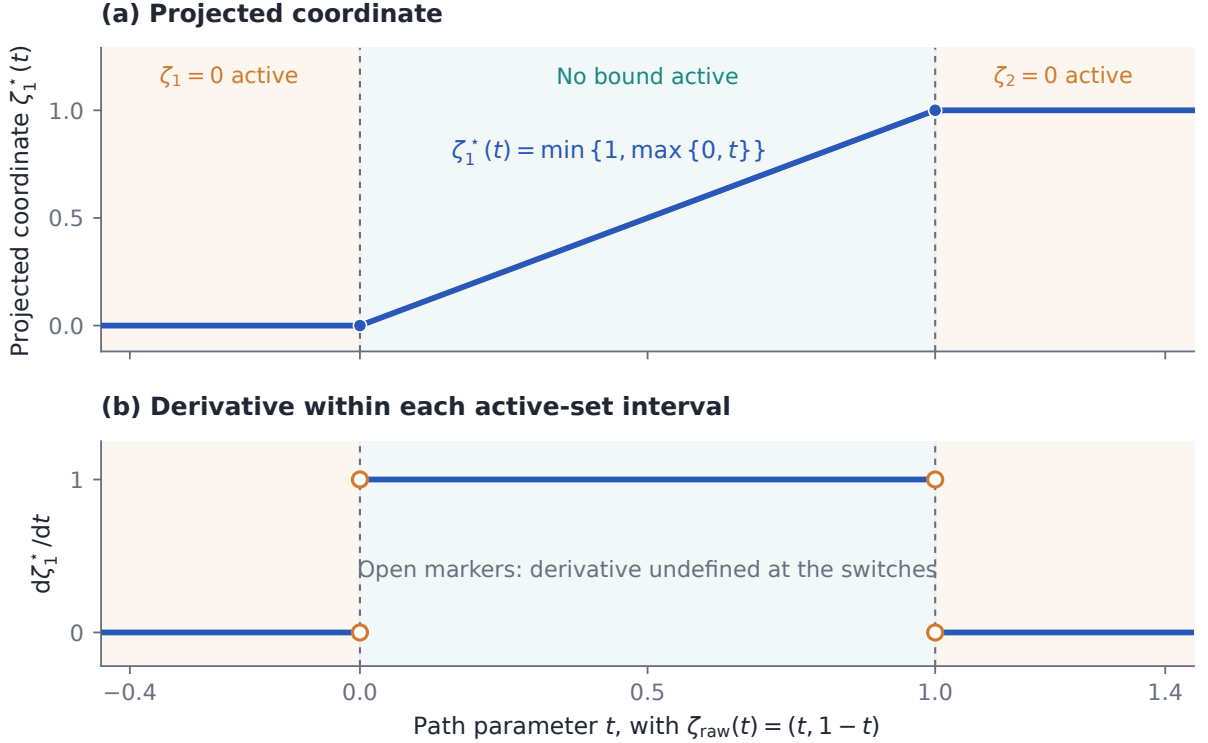


Figure 5: Active-set geometry of an analytical parametric projection. The response minimizes $\frac{1}{2}\|\zeta - \zeta_{\text{raw}}(t)\|_2^2$ over $\{\zeta \in \mathbb{R}_+^2 : \zeta_1 + \zeta_2 = 1\}$, with $\zeta_{\text{raw}}(t) = (t, 1 - t)$. (a) The solution is $\zeta_1^*(t) = \min\{1, \max\{0, t\}\}$ and $\zeta_2^* = 1 - \zeta_1^*$. (b) Its derivative is zero, one, and zero on the three open active-set intervals. At $t = 0$ and $t = 1$, the active bound has zero multiplier and unequal one-sided derivatives; open markers denote the undefined derivative.

2.8. Direct smooth mechanistic comparator

The surrogate route can now be compared with an optimization model that imposes the mechanistic balances directly. Route M uses the same operating controls and objective components, but replaces regression and projection with reactor and clarifier equations. The methodology distinguishes native normalization and the engineering rules applied to the retained decisions. Its state contains the concentrations that accumulate in the reduced mechanistic model as

$$y = \text{col}(c_1, \dots, c_N, s_1, \dots, s_L) \in \mathbb{R}_+^{n_y}, \quad n_y = NF + L. \quad (34)$$

The mixer and clarifier outlet vectors are reconstructed algebraically from (ϑ, x, y) . The objective and constraints use the reduced response with $M_{\text{cl}} = \sum_{\ell=1}^L V_{\text{cl},\ell} s_\ell$. This mapping allows both routes to evaluate the same engineering quantities despite their different state dimensions.

Outlet reconstruction divides by the clarifier-feed TSS, which could otherwise reach zero at an infeasible solver iterate. An auxiliary variable $\tilde{X}_F \geq X_F^{\min} > 0$ supplies a positive denominator, with the equality $\tilde{X}_F = t_X^\top c_N$ enforced

at feasibility. A differentiable residual $f_{\text{sm}}(\vartheta, x, y, \tilde{X}_F)$ contains all NF reactor balances and all L finite-volume clarifier balances. The direct route also retains the full layer envelope underlying the surrogate inventory bounds. Define $g_{\text{env}}(y) = \text{col}(s_1 \mathbf{1}_{L-2} - s_2; s_2; s_2; s_2; s_2 \mathbf{1}_{L-2})$. The auxiliary lower bound already imposes the feed-TSS engineering limit, so $g_{\text{eng,red}}$ omits that duplicate row from g_{eng} . The resulting nonlinear program (NLP) is

$$\begin{aligned} & \min_{\vartheta, y, \tilde{X}_F} J[\vartheta, \chi(\vartheta, x, y, \tilde{X}_F); \varpi] \\ & \text{subject to} \quad f_{\text{sm}}(\vartheta, x, y, \tilde{X}_F) = 0, \\ & \quad \tilde{X}_F = t_X^\top c_N, \quad \tilde{X}_F \geq X_F^{\min}, \\ & \quad \vartheta \in \mathcal{V}, \quad y \geq 0, \\ & \quad g_{\text{env}}(y) \leq 0, \\ & \quad g_{\text{eng,red}}[\vartheta, \chi(\vartheta, x, y, \tilde{X}_F)] \leq 0. \end{aligned} \tag{35}$$

Its selected feasible incumbent, when one exists, is denoted $(\vartheta_M, y_M, \tilde{X}_{F,M})$. The auxiliary does not change a feasible solution but keeps reconstruction defined during the search.

Derivative-based NLP methods use derivatives of the objective and constraint functions (Buzzi-Ferraris and Manenti, 2013). Here, smooth approximations make the kinetic and settling equations differentiable for optimization. They are used only for minimum, maximum, positive-part, zero-denominator, and receiver-switch operations. A nonzero smoothing width changes the equations near a switch. The smooth endpoint must therefore be checked on the original nonsmooth model before its performance is compared with the surrogate-selected decision. Both routes use that same reference model for evaluation.

The reference model evaluates a selected decision by holding its controls and influent fixed and solving the original reactor and clarifier equations to steady state. This calculation checks the resulting treatment performance independently of the surrogate prediction or the smoothed optimization equations. The reference model also has a defined physical scope. Its scalar-layer clarifier tracks aggregate solids, and particulate outlet components are reconstructed with the steady feed composition. Time integration uses a backward differentiation formula (BDF) for stiff equations (Curtiss and Hirschfelder, 1952). In this study, integration provides a numerical route to the nonsmooth steady equations. The model does not track the transient inventory of every particulate species in every layer. The reference response is an independently converged solution of those nonsmoothed equations. Stability refers only to this reduced numerical state model. The methodology below specifies how data from this model are used to fit the surrogate and how both routes are evaluated on the common reference.

3. Methodology

The evaluation follows seven ordered steps. First, the mechanistic model generates accepted plant states over the operating and influent domains. Second, independent streams provide model-development and post-selection holdout sets. Third, grouped cross-validation selects the raw-response and overflow-solids regressions. Fourth, out-of-fold predictions are projected and used to set trust limits. Fifth, the surrogate and direct mechanistic routes select operating decisions. Sixth, both decisions are evaluated on the same nonsmooth mechanistic model. Seventh, eligible pairs are compared by reference-model objective, controls, and optimization time. Prediction assessment, decision evaluation, and paired comparison therefore remain separate.

3.1. Case-study system and operating domain

The case study uses Activated Sludge Model No. 2d with two-step nitrification (ASM2d-TSN). It has $F = 20$ components, $F_S = F_X = 10$, and $K = 5$ named invariants. Five equal-volume reactors are followed by a ten-layer clarifier. Reactors 1 and 2 are unaerated. Reactors 3–5 have independent aeration controls. The seven free operating coordinates are

$$\vartheta = (H, a_3, a_4, a_5, r_I, r_R, w)^\top \in \mathbb{R}^7, \tag{36}$$

where $h_i = H/5$, $a_1 = a_2 = 0$, $k_L a_i = 47 a_i \text{ d}^{-1}$ for $i = 3, 4, 5$, and $\xi_i = 0$ for every stage.

Five equal-volume stages provide a compact anoxic–aerobic train for the comparative calculation. Ten 0.4-m clarifier layers resolve the specified 4-m depth without making the layer count a structural constant. The operating box spans

Table 2

Case-study topology, geometry, and free-coordinate bounds.

Quantity	Value or range	Case-study role
Q_0	10,000 m ³ d ⁻¹	Fresh-flow basis
N ; L ; ℓ_F	5; 10; 5	Reactors; layers; feed layer
f_i^V	1/5	Equal reactor-volume fractions
A_{cl} ; depth	1500 m ² ; 4.0 m	Clarifier geometry
H	6–36 h	Total nominal HRT
a_3, a_4, a_5	independently 0–1	Stage aeration settings
r_I	0–4	MLR ratio
r_R	0.25–1.25	RAS ratio
w	0.001–0.05	WAS ratio

low-to-high residence time, aeration, recycle, and wasting settings while preserving $q_E > 0$ and $q_U > 0$. These choices define the case-study domain and are not universal design bounds.

For this case, extended ICSOR uses 27 inputs, 406 features, and 161 response coordinates. The mechanistic route has 110 state coordinates. The projection imposes 76 equalities and 12 physical inequalities. Response non-negativity increases the total inequality count to 173. These dimensions follow from the declared plant configuration and are not fixed properties of the method.

The objective gives the largest individual weight to effluent quality while retaining explicit penalties for resource use. Its priorities are

$$\varpi = (0.50, 0.15, 0.20, 0.05, 0.05, 0.05)^\top, \quad \varpi_Q = \frac{1}{4}\mathbf{1}_4. \quad (37)$$

Aeration receives the largest resource coefficient because oxygen transfer is an important energy demand in the treatment train (Leu et al., 2012). HRT represents installed biological volume. The remaining resource coefficients apply to MLR, RAS, and wasted-solids production. The coefficients are trade-off parameters rather than percentages of the attained objective. The direct route uses $D_{T,M} = \text{diag}(\sigma_j)$ for positive effluent standard deviations computed from the model-development set. The surrogate route uses $D_{T,S} = \text{diag}[\max(1, \sigma_j)]$. Both reference-model objectives use $D_{T,M}$. The fixed aeration normalization is $E_A^{\text{ref}} = \sum_{i=3}^5 V_i(H^U)(47 \text{ d}^{-1}) = 423,000 \text{ m}^3 \text{ d}^{-1}$.

Both optimization routes use the same case-specific engineering constraints. These constraints restrict underflow TSS, prevent a nearly solids-free clarifier feed, and require positive external solids loss. Their complete definitions and values are provided in the Supplementary Material, Section S5. The wasted-solids objective uses $X_U^{\text{ref}} = 15,000 \text{ g TSS m}^{-3}$. The complete influent domain, reporting map, model constants, initial states, and numerical generation settings are provided in the Supplementary Material, Sections S1 and S2.

3.2. Mechanistic generation and analysis sample

The mechanistic design spans the joint seven-control and twenty-influent domain through Latin-hypercube sampling (McKay et al., 1979). Every candidate is solved from two declared initial states on the nonsmooth five-reactor and ten-layer model. A state is accepted only when both starts give finite and non-negative states and pass conservation, domain, layer-residual, stability, and branch-agreement checks. The full state is then mapped deterministically to the 161-coordinate response used for fitting.

The analysis attempts 10,000 mechanistic candidates. The development design contains 8,000 candidates from one seeded Latin hypercube. The separate holdout design contains 2,000 candidates from another seed. Candidates that pass the mechanistic acceptance checks form the corresponding analysis sets. Rejected candidates remain in the audit record and are not replaced.

The retained sets are conditioned on numerical solvability and may underrepresent regions where mechanistic checks fail. The holdout stream is separate from the model-development stream, but it is not independent external confirmation. Separating model development from prediction assessment reduces leakage of information into reported predictive performance (Kaufman et al., 2012). The response definition, overflow closure, numerical procedures, and engineering assessment rules were informed by inspection of the simulation data. The holdout and scenario results

are therefore described as post-selection evidence. Holdout and optimization responses do not alter the fitted models, preprocessing, trust limits, objective priorities, engineering constraints, or solver settings.

3.3. Model fitting, projection assessment, and trust calibration

All responses from one simulated plant remain in the same fold. The accepted model-development states are divided reproducibly into five folds whose sizes differ by at most one. Centers and scales are recomputed within each fitting fold. The raw multiresponse regression uses ridge regularization with an unpenalized intercept. Its score is the standardized root mean square error (RMSE) over the complete raw response. The one-standard-error rule selects the largest penalty whose mean score is within one standard error of the minimum (Hastie et al., 2009). Projection metrics do not influence this selection.

The scalar logarithmic overflow-solids model uses the same folds, preprocessing, penalty grid, and selection rule. Its score is calculated on the logarithmic target. After selection, both models are refitted on all accepted model-development states. Out-of-fold predictions are used for projection assessment and trust calibration so that these calculations do not reuse fitted values from the same rows (Hastie et al., 2009). Final fits are used only for the holdout, optimization, and selected-decision evaluations.

Every projected response is checked independently for equality feasibility, inequality feasibility, non-negativity, dual feasibility, stationarity, and complementarity. Scientific admission requires finite and audited out-of-fold projections for all model-development states. It also requires raw normalized RMSE below one for the complete response and the clarifier-inventory coordinate. Because each error is divided by its model-development response scale, this cutoff requires aggregate error below one response-scale unit. It is a study-specific admission rule rather than a universal accuracy threshold. A failed gate does not trigger refitting.

The trust region uses four diagnostics. They measure standardized projection correction, regularized feature leverage, particulate-recovery deviation, and smooth reactor-balance residual. Correction, recovery, and reactor limits are the out-of-fold 95th percentiles from the model-development set. These quantiles retain the range observed for most development rows while screening the upper tail. The leverage limit is the maximum model-development leverage under the final feature map, so it covers the observed development range. These limits are empirical guardrails rather than probabilistic confidence bounds. All four limits are fixed before optimization. The post-selection holdout reports raw and projected RMSE, mean absolute error (MAE), bias, normalized root mean square error (nRMSE), normalized mean absolute error (nMAE), and coefficient of determination (R^2) for each process block and the complete response. Overflow TSS is also assessed on logarithmic and concentration scales. The holdout figures calculate nRMSE and nMAE for COD, TN, TP, and TSS after dividing each prediction error by the mechanistic holdout range of the same composite at the same location. Aggregate metrics pool these normalized errors. The reported R^2 is the arithmetic mean of the location-composite coefficients of determination. These reporting scales differ from the development scales used in fitting and admission checks. The detailed fitting equations, sampling seeds, projection certificates, error bounds, scales, and tolerances are provided in the Supplementary Material, Sections S3 and S4.

3.4. Operating searches and convergence assessment

The nominal influent is the midpoint of each declared influent interval. Both routes use the seven controls in Equation (36), the bounds in Table 2, the objective priorities in Equation (37), and the same engineering constraints. The principal comparison initializes both searches at the center of the normalized operating box. Alternative mechanistic initializations are assessed separately in Section 3.7.

The surrogate search evaluates only the seven controls. At each distinct trial, the projection QP is solved without reusing an earlier trial's solution and receives a complete independent audit (Stellato et al., 2020; Boyd and Vandenberghe, 2004). Response derivatives are obtained from the active-set Karush–Kuhn–Tucker (KKT) system when rank, conditioning, complementarity, and derivative checks pass (Tøndel et al., 2003). Sequential least squares programming (SLSQP) then performs the outer search (Kraft, 1988). If the derivative chain cannot be validated, Constrained Optimization BY Quadratic Approximations (COBYQA) continues with function values from the same audited projection (Ragonneau, 2022).

A surrogate candidate receives first-order stationarity evidence only when the derivative, projection, and upper-level KKT audits pass. Otherwise, complete feasible-direction polls at two declared resolutions test for sufficient decrease. These finite polls provide evidence only for the tested directions and resolutions. They do not establish stationarity or local optimality because direct-search stationarity results require limiting behavior and suitable refining directions (Audet and Dennis, 2006). Neither convergence tier establishes global optimality.

The direct route uses the Interior Point OPTimizer (IPOPT) (Wächter and Biegler, 2006). The study applies three smoothing stages and a full-state endpoint audit. If the initial direct solve fails, one additional solve may start from the same scenario's certified surrogate controls and repeat the same smoothing sequence. This recovery depends on route S and is not an independent search. A validated feasible candidate remains reportable when a search budget is exhausted or stationarity is unresolved. A scenario without a validated candidate remains a failure. The full solver settings, poll construction, smoothing definitions, and recovery rules are provided in the Supplementary Material, Section S5.

3.5. Mechanistic evaluation and paired decision comparison

Native objective values mix decision quality with response error and route-specific normalization, as shown by Equation (38). Each available selected candidate $k \in \{S, M\}$ is therefore evaluated from two initial states on the nonsmooth ten-layer mechanistic model. This higher-fidelity check follows the validation role used in surrogate-based optimization (Bhosekar and Ierapetritou, 2018). The evaluation must pass state agreement, mass conservation, non-negativity, layer residual, domain, reduced-stability, and branch checks. Boundary-ambiguous branch disagreement may be qualified only when state agreement and all physical checks pass. The full evaluation state is mapped to $\chi_{\text{ref},k}$ using Equations (14) and (12).

Native responses are $\chi_{\text{nat},S} = \hat{\chi}_S$ and $\chi_{\text{nat},M} = \chi_{\text{sm},M}$. Extended ICSOR is also evaluated, wherever defined, at both selected sets of controls. Let $J_{\text{nat},k}$ use the native normalization of route k and let J_{ref} use $D_{T,M}$. The objective comparisons are

$$\begin{aligned}\Delta J_{\text{nat-ref},k} &= J_{\text{nat},k}(\vartheta_k, \chi_{\text{nat},k}) - J_{\text{ref}}(\vartheta_k, \chi_{\text{ref},k}), \\ \Delta J_{S-M} &= J_{\text{ref}}(\vartheta_S, \chi_{\text{ref},S}) - J_{\text{ref}}(\vartheta_M, \chi_{\text{ref},M}).\end{aligned}\quad (38)$$

Each valid, physically audited evaluation retains its objective and components. The reference-model objectives are compared on a common basis when both routes provide feasible candidates and valid mechanistic evaluations. Detailed reporting and comparison rules are provided in the Supplementary Material, Section S6. The objective difference is not termed regret because neither route has a global optimality certificate. Symmetric relative objective difference and normalized control root mean square (RMS) and maximum differences accompany each comparison. These casewise checks do not establish smooth and nonsmooth model equivalence over the full operating domain.

3.6. Influent scenarios, optimization time, and failure accounting

Ten influent vectors from a separate 20-dimensional Latin hypercube define the post-selection influent scenarios. They do not enter fitting, calibration, or tuning. Both routes are attempted for the nominal influent and for every influent scenario. Reference evaluation is performed whenever a selected decision is available. The recorded outputs include route status, controls, treatment responses, clarifier inventory and layer profiles, engineering measures, objective components, projection correction, and optimization time. These cases show sensitivity within the evaluated set and do not establish population-level reliability.

Optimization time measures the duration of each route's optimization search. It includes projection evaluations and any derivative-free fallback for Extended ICSOR, and the smoothing sequence for Smooth NLP. Data generation, fitting, convergence certification, additional recovery calculations, and evaluation with the original mechanistic model are excluded. Aggregate time comparisons use the ten influent scenarios and exclude the nominal scenario. Both routes use the same processor allocation and thread count. Workload and failure records accompany the optimization time summaries. This study does not accept a candidate from the solver success flag alone. The candidate must also pass the applicable numerical, physical, stability, and branch checks. Detailed metric definitions, evaluation tolerances, workload fields, and failure rules are provided in the Supplementary Material, Sections S5–S7.

3.7. Sensitivity analyses and computational assessment

The sensitivity analyses use the same accepted development and holdout states and the same grouped cross-validation folds as the quadratic surrogate. Fitting, scale estimation, predictor selection, and diagnostic calibration use only development data. The holdout provides a common set of inputs for descriptive prediction comparisons. Mechanistic evaluations at selected operating controls are decision checks and remain separate from the fitting and holdout sets. Detailed assessment settings are given in the Supplementary Material, Section S9.

3.7.1. *Projection constraints and overflow closure*

The constraint comparison includes the full projection, removal of componentwise densification, removal of inventory bounds, removal of the overflow closure, and a basic conservation projection. All variants retain the response coordinates, raw predictor, response metric, and applicable constraint-row scales. Prediction errors are compared on matched inputs for which both projections pass their numerical audits. Acceptance rates use all attempted inputs, including failed projections. Overflow TSS and clarifier inventory receive separate error summaries. For operating-decision sensitivity, the full, no-densification, and no-inventory variants use the same value-based optimizer, initial controls, evaluation budget, objective, engineering limits, and influent scenarios. Selected endpoints undergo an independently initialized projection and mechanistic evaluation. Their feasibility and stationarity qualifications are reported separately.

Closure sensitivity varies the regularization of the scalar logarithmic regression while retaining the fitted raw plant predictor. Each closure is assessed on the holdout and on an independent Latin hypercube spanning the declared operating and influent domain. The latter assessment measures projection acceptance without requiring mechanistic targets at every sampled input. An independent linear-feasibility calculation diagnoses rejected projections; closure incompatibility requires a feasible witness after removal of the closure equality alone at the same input. This distinction separates numerically diagnosed incompatibility from unresolved solver failure. The sampled acceptance rates describe the evaluated inputs and provide no guarantee of feasibility throughout the continuous domain.

3.7.2. *Predictor comparison and diagnostic screening*

The Gaussian-kernel predictor uses fitting-fold preprocessing and development-only landmark selection and tuning. Its raw and projected responses are evaluated on the same holdout as quadratic regression, with the common physical projection and overflow closure. This comparison assesses the dependence of prediction error and projection acceptance on the statistical approximation.

Trust sensitivity multiplies the four development-calibrated diagnostic limits by fixed factors of 0.5, 1, and 2. The factors apply to the holdout screens and scenario operating searches without recalibration. Screening is assessed separately against high standardized prediction error and failure of the retained engineering and physical checks in the mechanistic reference. The high-error cutoff is the nearest-rank 95th percentile of complete-response errors from accepted development out-of-fold projections. Error labels require finite, accepted projected responses; inputs without such labels retain their projection-failure classification. False-screening and missed-screening rates use the corresponding reference-class denominators. Since the holdout contains only states accepted by the mechanistic generation checks, these rates characterize prediction error and engineering feasibility within that accepted population.

3.7.3. *Projection cost and mechanistic initialization*

Projection timing compares independent initialization with initialization from the preceding accepted primal–dual solution on identical ordered inputs. The inputs comprise holdout samples and sampled chronological sequences from the optimization searches, with initialization reset between sequences. The warm-initialized path includes an independently initialized replay and numerical audit. Measurements distinguish preliminary solver setup and solution from total projection-call time, including replay and audit. Regression evaluation and network-operator assembly precede these intervals. Iteration counts, retries, acceptance, and standardized replay disagreement accompany the timings. The solver is rebuilt for each input, so the comparison measures iterate initialization without persistent factorization reuse. Active-set derivative availability is reported relative to attempted derivative evaluations, with rejection reasons distinguished from projection failure.

Mechanistic initialization compares three routes: box-center controls with the nearest-development-state initializer; surrogate-selected controls with the same state initializer; and surrogate-selected controls with an independently accepted mechanistic seed state. The third route requires a valid two-start mechanistic evaluation of the seed. All routes use the same objective scales, continuation sequence, per-stage solver budgets, and engineering constraints. Surrogate search, seed-state evaluation, direct solution, and final mechanistic evaluation contribute separately to the total cost. Reference-model objective and engineering feasibility accompany the timing comparison, and unsuccessful attempts remain in its denominator.

3.7.4. *Cost under repeated use*

Computational accounting separates mechanistic data generation, surrogate fitting, online search, convergence assessment, and reference-model evaluation. For repeated operating decisions, let C_S and C_M denote applicable one-time costs and let c_S and c_M denote mean per-decision costs measured over the same phases. The computational

crossover is

$$n_{\text{cross}} = \left\lceil \frac{C_S - C_M}{c_M - c_S} \right\rceil, \quad C_S > C_M, \quad c_M > c_S. \quad (39)$$

If either condition fails, the total-cost comparison determines whether a route dominates or no positive crossover exists. Three allocations treat the simulation data as already available, charge development-data generation to the surrogate, or additionally charge holdout-data generation. The measured offline costs comprise generation and regression fitting, including cross-validation. Trust calibration and model-design effort are outside those measurements. Online search costs use the timing boundary defined above, while hybrid-route totals include surrogate selection and mechanistic initialization. The crossover therefore describes computational cost under the stated allocation and timing boundary; decision quality remains a separate comparison on the common mechanistic reference.

4. Results and discussion

The results address the three research questions in sequence. The holdout assessment first examines how projection affects prediction accuracy across the connected plant. Selected operating decisions are then evaluated with the original nonsmooth mechanistic model to check their treatment performance. Finally, both routes are compared using the same reference model, objective priorities, and operating bounds. Optimization time is assessed separately. N denotes the nominal influent, and S1–S10 denote the ten influent scenarios. Numerical values in this section are calculated from the figure data and rounded only for presentation.

4.1. Holdout accuracy and projection

4.1.1. Aggregate prediction accuracy

The holdout assessment uses 1,845 accepted mechanistic states. Projection improves the aggregate metrics in Figure 6. nRMSE decreases from 0.04690 to 0.04306, and nMAE decreases from 0.03044 to 0.02568. The relative reductions calculated before rounding are 8.19% and 15.65%, respectively. Mean location-level R^2 increases from 0.925 to 0.938. Improvement occurs across all eight location summaries and all four composite summaries, although its magnitude varies.

Projection also enforces the stated physical relations. Each feasible, audited projected response satisfies the mixer balance, stagewise invariant relations, clarifier outlet balances, soluble pass-through, particulate densification, inventory bounds, and non-negativity requirements to the specified numerical tolerances. These constraints connect the predictions used to calculate treatment performance and resource measures.

The projection imposes only the stated relations. Biological kinetics and nonlinear settling equations remain outside its constraint set. The fitted overflow-TSS value can also differ from the mechanistic value. Lower prediction error at every coordinate is therefore not guaranteed (Selerio, 2026; Sturm and Silva, 2025). Figure 6 reports the observed accuracy of the regression, overflow model, and projection used together.

4.1.2. Location-specific effects of projection

The clarifier underflow has the largest reduction in location-level nRMSE, from 0.05814 to 0.04736. Across the mixer and reactors, nRMSE ranges from 0.04431 to 0.04466 before projection and from 0.04187 to 0.04245 after projection. The underflow supplies both returned biomass and wasted solids, so an error in this stream can affect recycle mixing and the solids-related objective terms (Zhang et al., 2026).

Figure 7 resolves these changes into 32 location–composite combinations. Projection lowers nRMSE in 30 of them. Overflow TP nRMSE decreases from 0.02948 to 0.01795, a reduction of 39.10%. Underflow nRMSE decreases for all four composites. The two increases occur in COD at R5 and the overflow, by 0.47% and 2.36%, respectively. These results describe the combined model and do not isolate the effect of individual projection constraints.

Overflow TSS has the highest projected nRMSE among the location–composite combinations, at 0.07392. Favorable average accuracy across the plant therefore does not establish equally strong prediction at the discharge point. This location requires direct assessment because its concentrations enter the water-quality objective.

4.1.3. All-location parity and residual structure

Figure 8 shows the distribution of predictions around the line of equal predicted and mechanistic concentrations. Projected nRMSE is 0.04586 for COD, 0.03454 for TN, 0.03349 for TP, and 0.05476 for TSS. The corresponding mean

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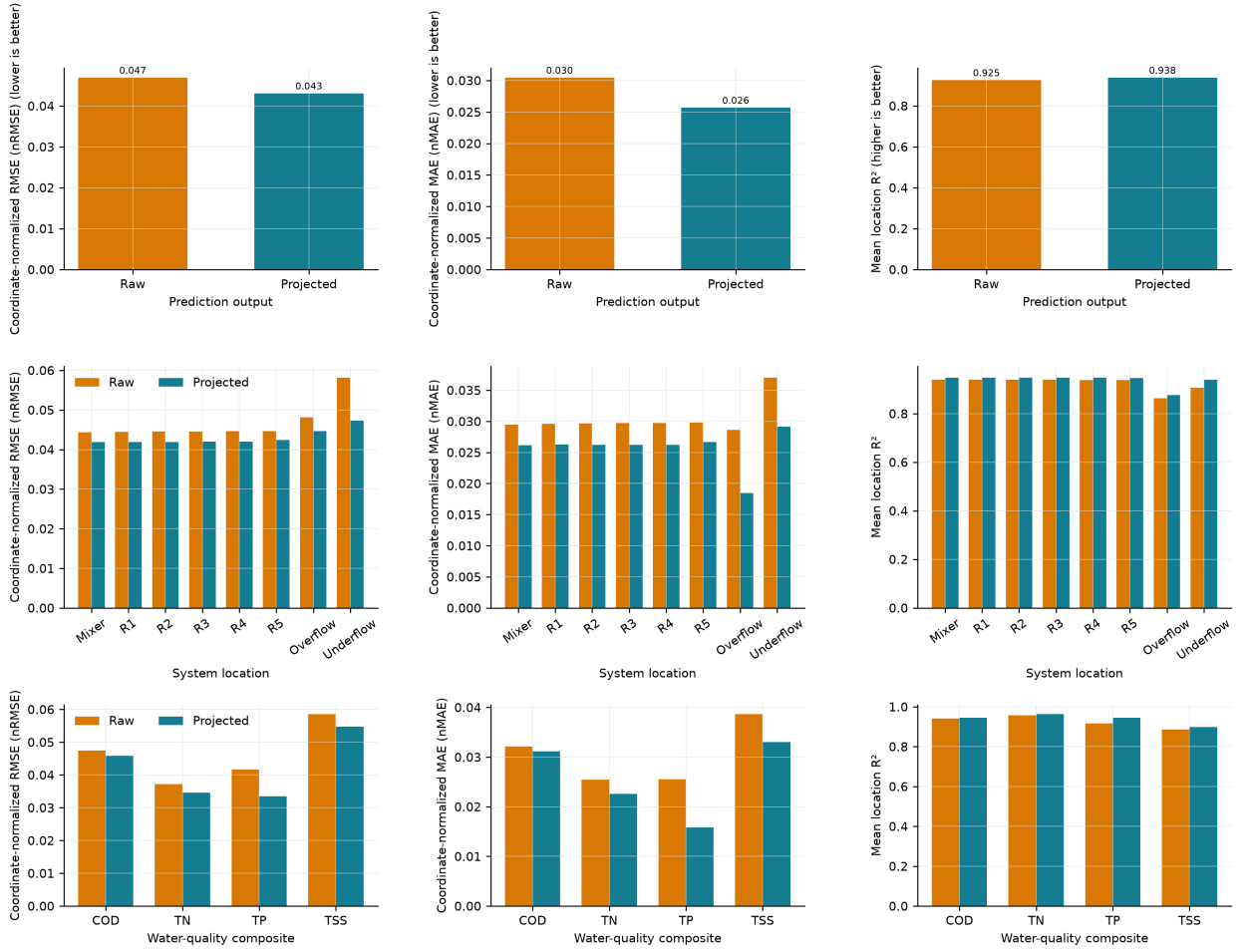


Figure 6: Raw and projected holdout accuracy. The top row presents aggregate coordinate-normalized RMSE, coordinate-normalized MAE, and mean location-level R^2 . The middle and bottom rows resolve these metrics by treatment location and water-quality composite, respectively.

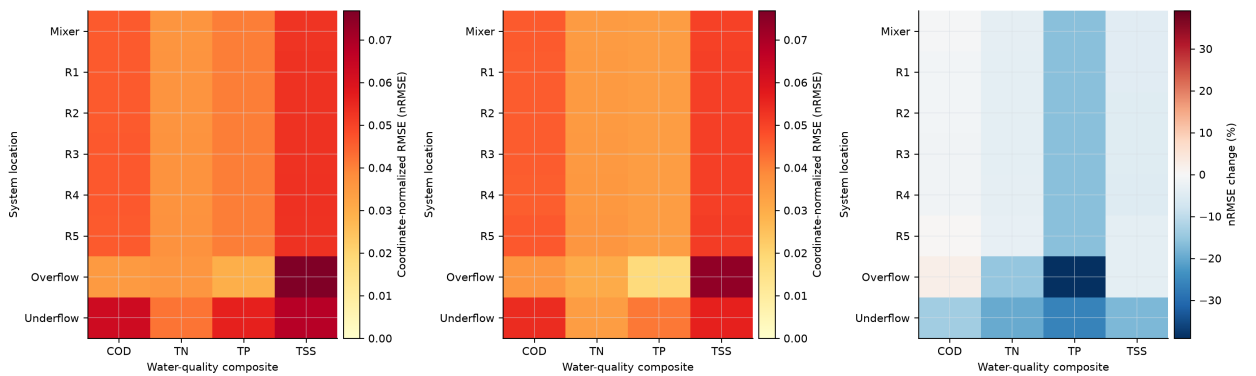


Figure 7: Holdout nRMSE by treatment location and reporting composite for raw predictions (left), projected predictions (middle), and their percentage change (right). Negative percentage changes indicate lower error after projection.

location-level R^2 values are 0.944, 0.963, 0.945, and 0.899. TSS has the largest normalized error and lowest mean R^2 of the four composites.

The hexagons summarize the density of all observations, while the labelled markers show the median at each location. Agreement near the central part of a concentration range can coexist with errors at its ends (Shahbazi et al., 2026). At the overflow, projected TSS is below the mechanistic value in 1,008 of the 1,845 holdout states. A positive overflow-TSS prediction therefore does not ensure a conservative estimate of the residual concentration.

Each holdout state contributes observations at eight connected locations. The hexagon counts consequently describe the distribution of location observations rather than independent plant replications. The logarithmic axes display a broad concentration range, while the coordinate normalization allows errors at different locations to contribute on comparable scales.

The holdout results show improved aggregate prediction accuracy after projection, with the largest location-level improvement at the underflow. Errors at the effluent remain relevant to operating decisions. Satisfaction of the imposed physical relations and accuracy against the mechanistic model must therefore be assessed separately (Selerio, 2026; Valente et al., 2025).

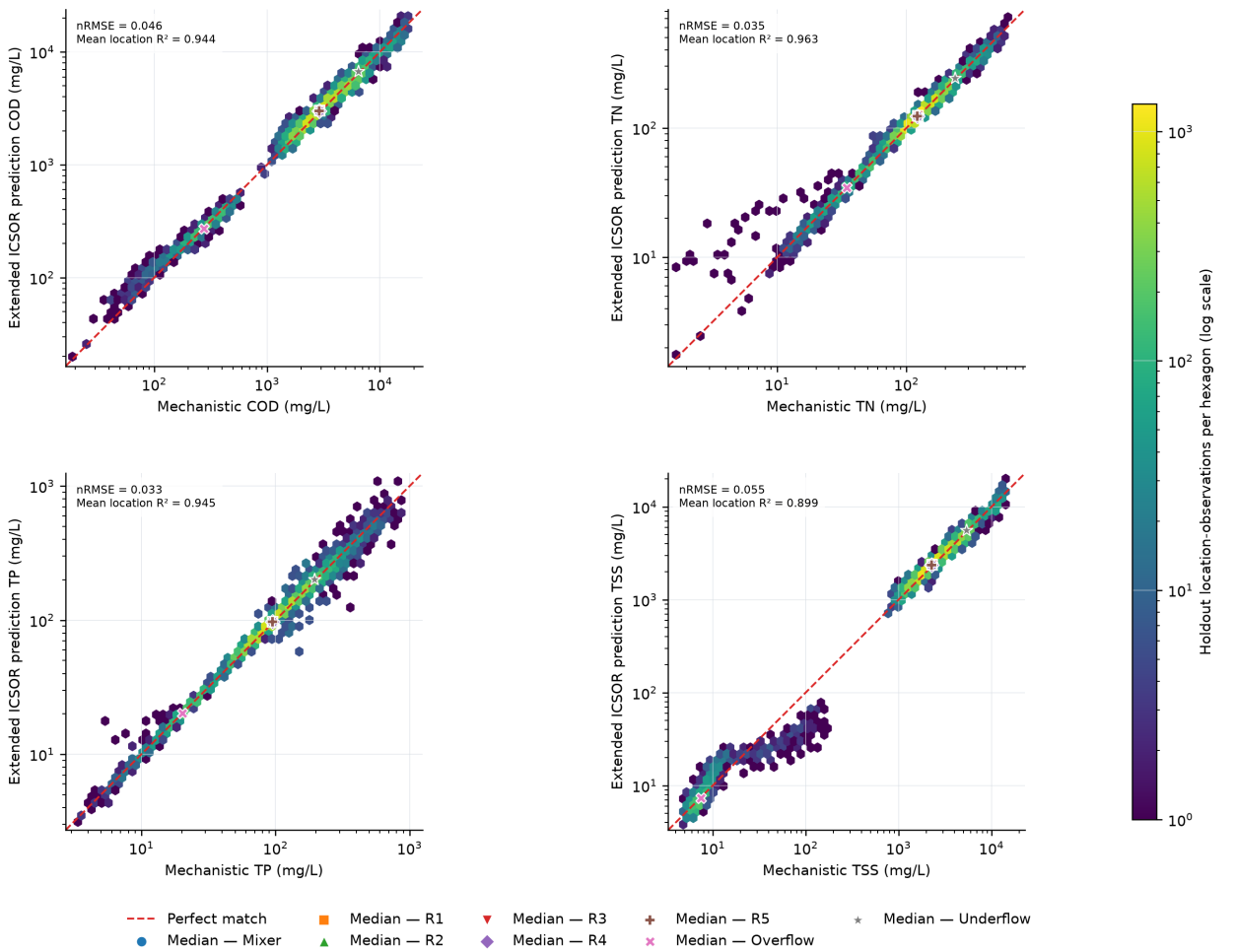


Figure 8: Projected Extended ICSOR parity for COD, TN, TP, and TSS across holdout rows and eight treatment locations. Both concentration axes are logarithmic. Hexagon colors indicate location-observation counts on a logarithmic scale. Markers identify location medians. Dashed lines denote perfect agreement.

4.2. Selected-decision fidelity and treatment behavior

4.2.1. Prediction fidelity at selected operating controls

Figure 9 evaluates prediction accuracy at the operating controls selected by Extended ICSOR. For each nominal or scenario decision, the controls and influent are held fixed while the original nonsmooth reactor and clarifier equations are solved to steady state from two initial states. This mechanistic evaluation provides the comparison concentrations and removals. It is distinct from solving the surrogate optimization problem. The plotted removal for composite j is $100(1 - z_{E,j}/z_{in,j})$, where $z_{in,j}$ is the fresh-influent concentration for that scenario.

Selected-decision COD errors vary across scenarios. In S6, Extended ICSOR predicts 151.10 mg L^{-1} , whereas the mechanistic evaluation gives 99.21 mg L^{-1} . In S10, the corresponding concentrations are 232.23 and 167.78 mg L^{-1} . These overpredictions of concentration underpredict removal by 8.51 and 10.44 percentage points, respectively. The errors are therefore not uniformly optimistic about treatment performance.

S4 illustrates the opposite direction of error for nutrients. Predicted TN is 2.67 mg L^{-1} and the mechanistic value is 7.47 mg L^{-1} . Predicted TP is 2.08 mg L^{-1} and the mechanistic value is 4.68 mg L^{-1} . The predictions overstate TN removal by 16.97 percentage points and TP removal by 27.33 percentage points. Such errors would affect assessment against a nutrient limit near these concentrations (Ren et al., 2026; Zhang et al., 2026).

Across the nominal and ten influent scenarios, predicted TSS ranges from 5.53 to 8.05 mg L^{-1} and mechanistic TSS ranges from 5.28 to 6.41 mg L^{-1} . TSS is overpredicted in ten of the eleven decisions. In S1, for example, the predicted and mechanistic concentrations are 7.78 and 5.77 mg L^{-1} . Their removal percentages differ by only 0.61 percentage points because both represent high removal relative to the influent. Reporting concentration and removal together exposes this difference in scale.

Projection maintains the enforced physical relations but does not eliminate prediction errors at selected decisions. The mechanistic evaluation is needed to establish the treatment performance used in the route comparison. This is particularly relevant when the predicted effluent approaches a prescribed treatment limit (Hameed et al., 2026; Ren et al., 2026).

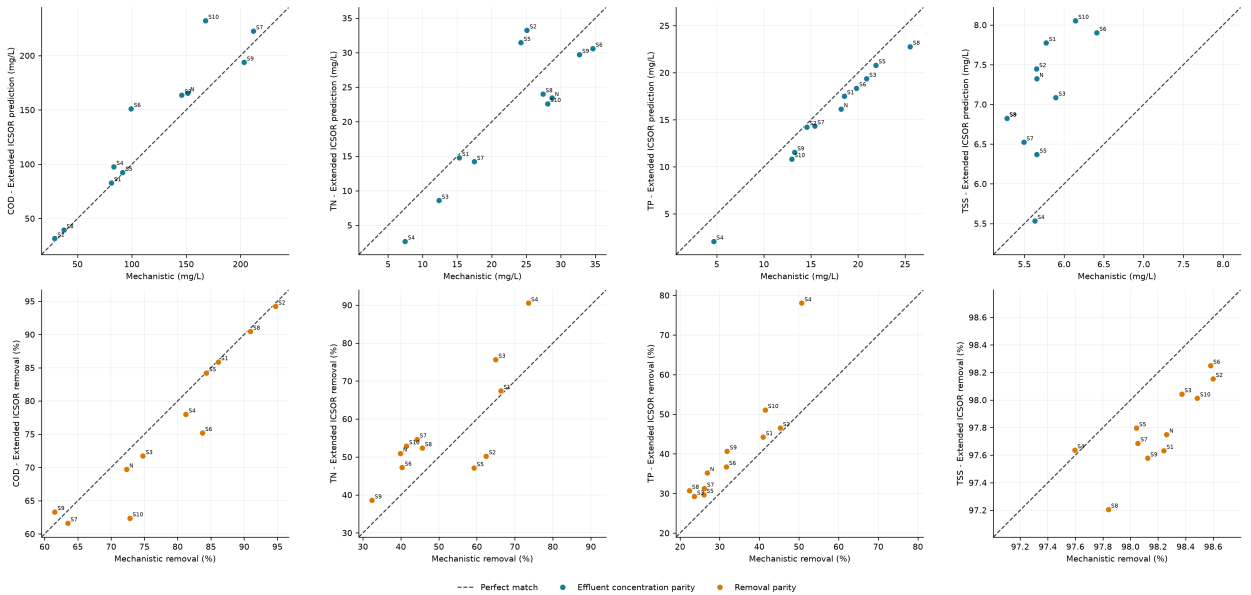


Figure 9: Extended ICSOR predictions compared with treatment outcomes calculated by the original nonsmooth mechanistic model at its selected controls. The upper row presents effluent concentration parity for COD, TN, TP, and TSS. The lower row presents parity of the reported removals. Teal points denote concentrations, orange points denote removals, and dashed lines denote perfect agreement. N identifies the nominal case and S1–S10 identify the influent scenarios.

4.2.2. Effluent outcomes and operating choices

Figure 10 compares mechanistic treatment outcomes and selected controls for both routes. Smooth NLP gives lower COD, TN, and TP in the nominal scenario and nine of the ten influent scenarios. In the nominal scenario, COD is 151.26 mg L^{-1} at the Extended ICSOR decision and 57.60 mg L^{-1} at the Smooth NLP decision. In S9, the corresponding values are 203.24 and 19.08 mg L^{-1} . COD outcomes are closer in S1, S2, S5, and S8.

Mechanistic TSS across both routes ranges from 5.28 to 6.95 mg L^{-1} . Smooth NLP has lower TSS in six of the eleven comparisons. In S8, it has slightly higher concentrations of all four composites. The TN values are 27.41 mg L^{-1} for Extended ICSOR and 28.52 mg L^{-1} for Smooth NLP, and TP is 25.53 and 25.59 mg L^{-1} , respectively. A lower total objective therefore does not require every effluent concentration to decrease.

Extended ICSOR selects the 6-h HRT lower bound in the nominal scenario and S6, S7, S9, and S10. Smooth NLP selects the 36-h upper bound in S5 and S10, within the numerical tolerance. Under fixed fresh flow and equal reactor-volume fractions, increasing H increases biological reactor volume. These comparisons therefore include capacity as well as operating choices. An assessment of an existing plant would need to fix its installed volumes (Zhang et al., 2026).

The normalized aeration setting in R3 is at its upper bound for both routes in every plotted scenario, within the numerical tolerance. R4 and R5 also reach the upper bound in many decisions. Identical aeration settings can still give different oxygen-transfer capacity because reactor volume depends on H (Miederer et al., 2025). The resource comparison must account for both volume and aeration.

Both routes select the lower RAS bound, $r_R = 0.25$, within the numerical tolerance. Smooth NLP selects internal recycle ratios of 4.00 in S2 and S10 and 3.44 in S9. Its other displayed internal recycle ratios are below 1.3×10^{-9} and are zero at the reported precision. Extended ICSOR selects ratios from 0 to 0.804. The tested influents therefore do not yield a single internal recycle setting.

The actual train residence time per pass is H/q_P , where $q_P = 1 + r_I + r_R$. In S2, the Smooth NLP decision has $H = 31.40 \text{ h}$, $r_I = 4.00$, and $r_R = 0.25$. These values give $q_P = 5.25$ and a residence time per pass of 5.98 h. Nominal HRT and recycle flow must therefore be interpreted together.

Wasting often reaches its upper bound, but its objective contribution depends on wX_U . The waste ratio alone does not determine wasted-solids production. Whole-plant SRT additionally depends on reactor and clarifier inventories (Alex et al., 2008).

4.2.3. Treatment-train profiles

Figure 11 follows the mechanistic concentrations through the treatment train at each route's selected controls. Both columns use the original reference model, so differences arise from the selected controls and resulting states. Each composite uses the same logarithmic concentration limits in the two route panels.

All four composites increase from fresh influent to the mixer in the nominal scenario and all ten influent scenarios for both routes. Returned sludge and mixed liquor reintroduce material circulating within the plant (Zhang et al., 2026). Equation (2) accounts for this increase. Internal recycle streams cancel from the external plant balance, so a higher mixer concentration does not indicate creation of an external pollutant load.

The composite profiles combine soluble and particulate contributions. Their changes across R1–R5 cannot by themselves identify nitrification, denitrification, or phosphorus uptake. A nearly constant total can conceal redistribution among component forms. Higher mixed-liquor TP also need not imply higher effluent TP when phosphorus leaves the clarifier with retained solids (USEPA, 2009; Zhang et al., 2026).

The concentration decrease between R5 and clarifier effluent occurs through particulate separation because the clarifier is nonreactive in this model. The biological stages determine the soluble and particulate composition entering it (Zhang et al., 2026). The TSS decrease shows the role of separation in the final effluent response.

The connected profiles show how recycle mixing, biological conversion, and solids separation determine the final discharge. A process diagnosis would additionally require soluble nutrient species, DO, outlet solids flows, and whole-plant SRT (USEPA, 2009; Alex et al., 2008). Solving the complete layer model supplies the nonlinear clarifier checks that the aggregate surrogate inventory cannot provide (Takács et al., 1991; Zhang et al., 2026).

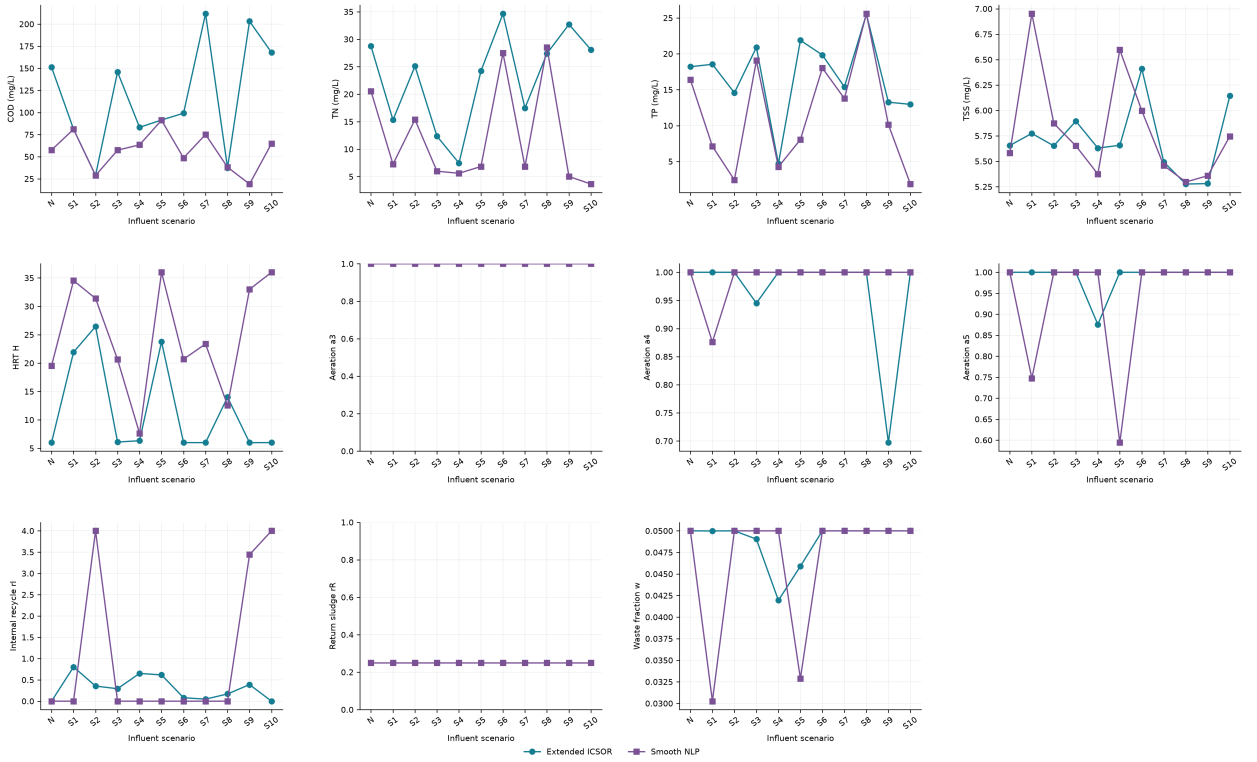


Figure 10: Evaluated effluent COD, TN, TP, and TSS and selected operating controls for Extended ICSOR and Smooth NLP. Controls comprise nominal HRT, aeration settings in R3–R5, internal recycle, return sludge, and wasting. N denotes the nominal case. S1–S10 denote the influent scenarios.

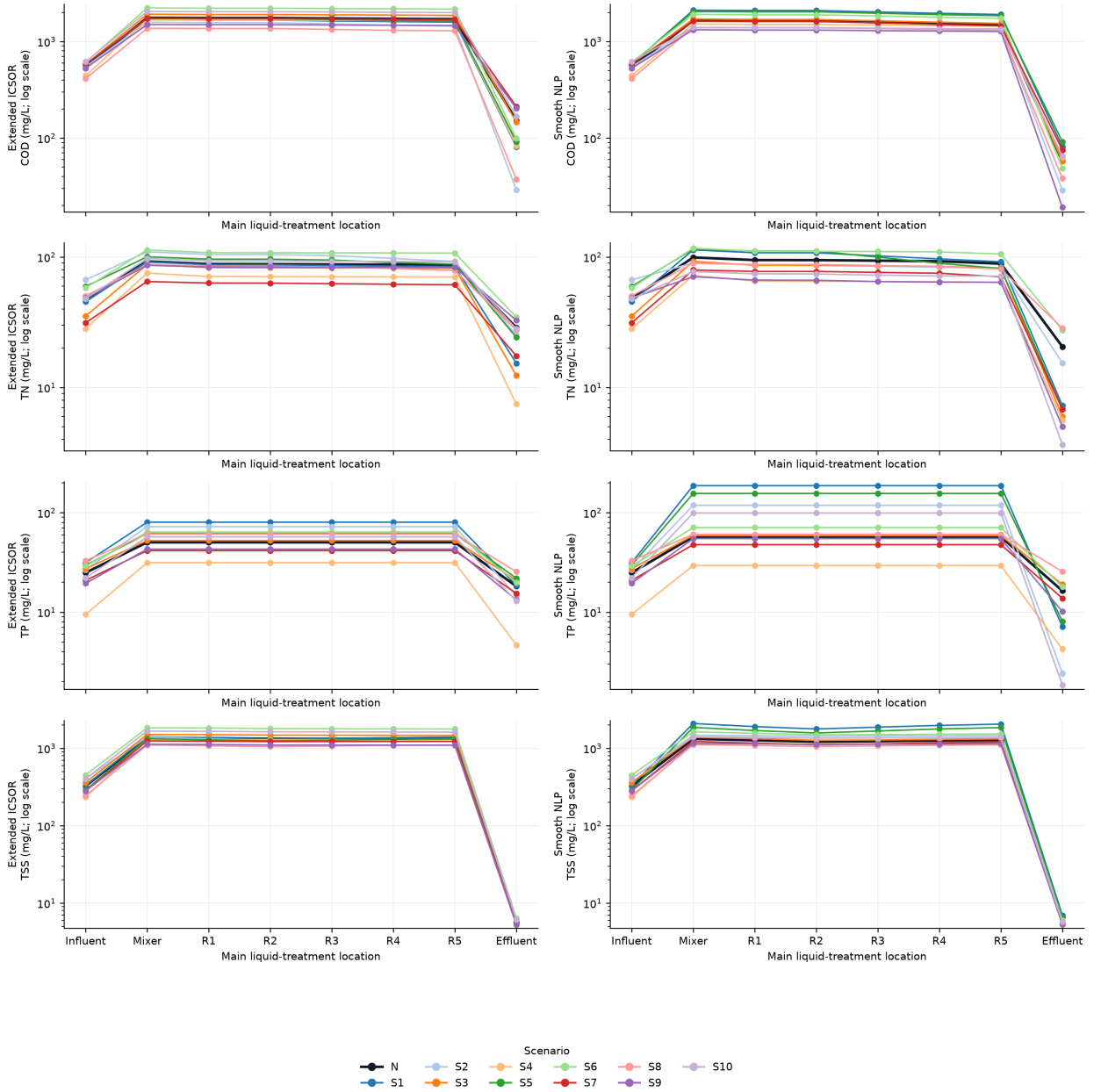


Figure 11: Concentration profiles calculated with the original mechanistic model at controls selected by Extended ICSOR (left) and Smooth NLP (right). Rows present COD, TN, TP, and TSS from fresh influent through the mixer and reactors R1–R5 to clarifier effluent. Concentration axes are logarithmic and share the same limits for both routes within each composite. Colors identify the nominal and influent scenarios.

4.3. Objective trade-offs and computational performance

4.3.1. Objective evaluated with the original mechanistic model

Smooth NLP has the lower reference-model total objective in the nominal scenario and all ten influent scenarios in Figure 12. The nominal objective is 0.8647 for Extended ICSOR and 0.7862 for Smooth NLP. The largest difference occurs in S5, with values of 1.0101 and 0.7022, respectively. In S6 and S8, the corresponding differences are 0.0117 and 0.0045. These comparisons describe the retained decisions under the stated priorities.

Smooth NLP has a lower normalized quality component in the nominal scenario and nine influent scenarios. Its weighted resource contribution is higher in every scenario except S8. In S8, the quality component increases from 1.5856 to 1.6110, while the resource contribution decreases from 0.1351 to 0.1179. The total objective consequently decreases from 0.9279 to 0.9234.

In S10, the normalized quality component is 1.4748 for Extended ICSOR and 0.3699 for Smooth NLP. Multiplication by the quality weight of 0.50 gives a reduction of 0.5525 in the weighted quality contribution. The weighted resource contribution increases from 0.0557 to 0.4191. The resulting total objective decreases from 0.7931 to 0.6040. This decision improves the weighted treatment outcome while accepting a larger capacity and resource penalty.

The aeration contribution requires particular care. For the equal-volume case study, its definition reduces to

$$j_A = \frac{H}{36 \text{ h}} \frac{a_3 + a_4 + a_5}{3}. \quad (40)$$

At $H = 6 \text{ h}$ and $a_3 = a_4 = a_5 = 1$, the aeration proxy is only one-sixth of its maximum-volume reference value. The frequent low-HRT Extended ICSOR decisions therefore receive small volume and aeration penalties even when their normalized aeration settings are maximal. This explains why control settings alone give an incomplete account of the resource comparison. The proxy measures oxygen-transfer capacity. Actual oxygen transfer also depends on the DO deficit (Zhang et al., 2026).

The native quality scales coincide in this dataset. The development standard deviations are 106.61, 13.07, 7.11, and 17.38 mg L^{-1} for COD, TN, TP, and TSS, respectively. All exceed the unit floor in $D_{T,S}$, so $D_{T,S} = D_{T,M}$. The route differences therefore cannot be attributed to that floor here. Response approximation, surrogate trust restrictions, and local search behavior can still affect the selected controls (Pedrozo et al., 2025; Hameed et al., 2026), and the comparison does not isolate their individual effects.

The objective combines engineering proxies under the priorities in Equation (37). Its reduction does not directly quantify monetary savings or measured energy consumption (Miederer et al., 2025; Zhang et al., 2026). Individual effluent concentrations still require assessment when prescribed treatment targets govern operation (Ren et al., 2026; Zhang et al., 2026).

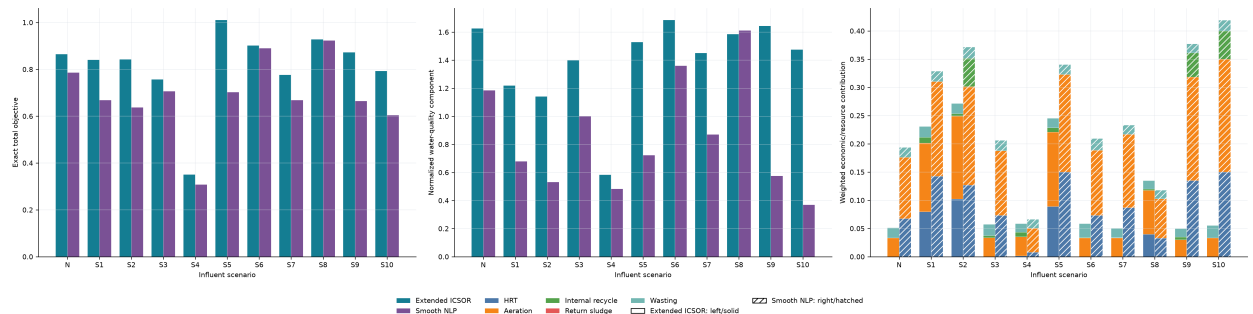


Figure 12: Reference-model total objective (left), normalized quality component j_Q (middle), and weighted resource contributions (right). Both routes use $D_{T,M}$ for reference evaluation. Resource stacks represent HRT, aeration, internal recycle, return sludge, and wasting. Extended ICSOR bars are solid and Smooth NLP bars are hatched. The quality panel displays j_Q , whose contribution to the total objective is $0.50j_Q$.

4.3.2. Optimization time

Extended ICSOR has the shorter optimization time in the nominal scenario and every influent scenario in Figure 13. Across S1–S10, its recorded times range from 0.758 to 2.702 s, compared with 3.932 to 8.253 s for Smooth NLP. The

means are 1.726 and 5.236 s, respectively. Nominal times are 0.985 and 5.062 s. The scenario ratios of Smooth NLP time to Extended ICSOR time range from 1.90 to 7.99, with the largest ratio in S10.

The surrogate search varies seven controls and obtains the response through regression and projection. Smooth NLP optimizes the controls together with mechanistic states through the smoothing sequence. These differences are consistent with the observed times, but the timing comparison alone does not identify which calculation dominates either search.

Both routes complete the measured optimization within seconds. Extended ICSOR uses less optimization time, whereas the retained Smooth NLP decisions have lower reference-model objectives. This result separates computational efficiency from the quality of the selected decisions.

The time and objective panels report different aspects of performance. Together they show faster surrogate optimization and lower reference-model objectives for the retained Smooth NLP decisions. They do not establish equal decision quality or a reduction in the time of the complete calculation (Bliek et al., 2023).

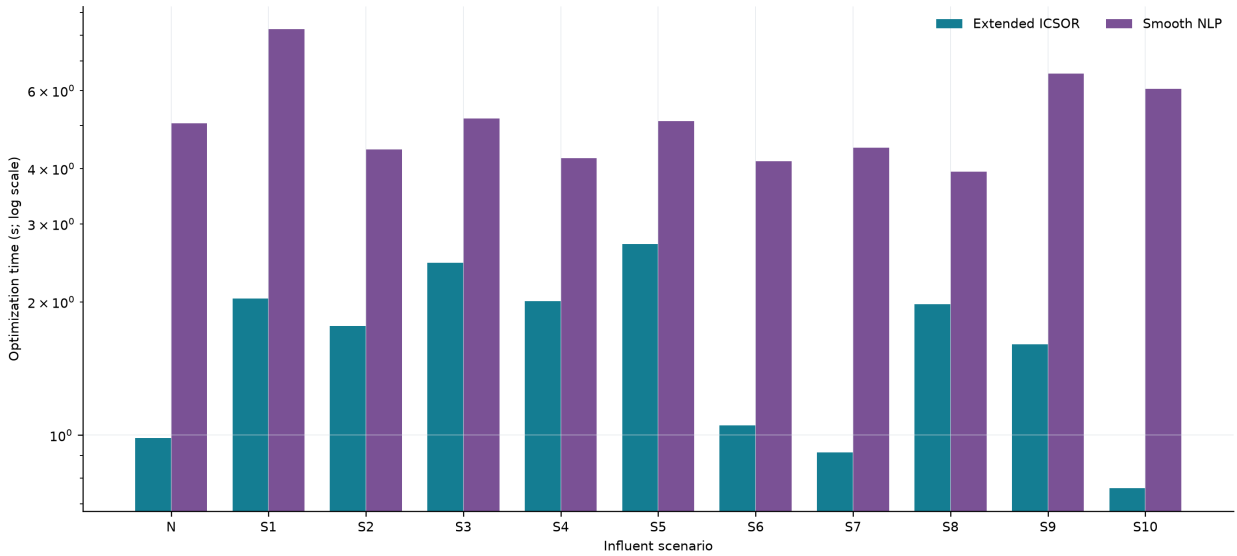


Figure 13: Optimization time for Extended ICSOR and Smooth NLP for the nominal influent and ten influent scenarios. The seconds axis is logarithmic. Optimization time excludes data generation, fitting, certification, conditional recovery, and evaluation with the original mechanistic model.

5. Conclusions

This study developed a plant-wide, physically constrained statistical surrogate for operational optimization of activated sludge systems. The model combines second-order ridge regression with joint quadratic projection. It represents the mixer, five-reactor train, clarifier outlets, and stored solids as one connected response. The projection enforces the stated mixer and clarifier balances, stagewise invariant relations, particulate densification, clarifier inventory bounds, and non-negativity. Separate model-development and holdout designs were used to assess prediction accuracy. Operating decisions were selected for the nominal influent and ten additional influent scenarios. Each retained decision was independently evaluated with the original mechanistic model and compared with the retained decision from Smooth NLP.

The results show that plant-wide physical projection can improve the statistical representation of the connected treatment system while enforcing the imposed physical relations. Among the 1,845 accepted holdout states, projection reduced aggregate nRMSE by 8.19% and nMAE by 15.65%. It also improved 30 of the 32 location–composite nRMSE values. The improvement was not limited to one treatment location or one water-quality measure. These results show that enforcing connected plant-wide relations can improve aggregate prediction accuracy rather than serving only as a feasibility correction.

The optimization results also show that physical consistency of surrogate predictions and quality of the resulting operating decisions are separate requirements. Mechanistic evaluation identified errors in selected nutrient and solids predictions that can affect the assessment of treatment performance. Smooth NLP produced a lower reference-model objective for the nominal influent and all ten influent scenarios. Extended ICSOR required less optimization time in every case. Across the ten influent scenarios, the mean optimization times were 1.726 s for Extended ICSOR and 5.236 s for Smooth NLP. The results therefore identify a clear trade-off in the tested cases. The surrogate route reduced the time spent in the optimization search, whereas the direct mechanistic route selected decisions with better reference-model objective values. Faster surrogate optimization should consequently not be interpreted as evidence of equal decision quality.

The main contribution is a statistical surrogate that represents a connected activated sludge plant and imposes its specified physical relations through one joint projection. The original ICSOR formulation provides the basis for this model. The plant response includes clarifier outlet flows and aggregate solids inventory without predicting every internal clarifier layer. This formulation supplies the quantities needed for the treatment objective and engineering constraints. The comparison shows why physical feasibility, prediction accuracy, decision quality, and computational cost require separate assessment. A physically consistent surrogate prediction with favorable average error can still lead to a decision with a higher objective when evaluated by the same mechanistic reference. This distinction provides a practical basis for assessing constrained surrogates used in process optimization.

The findings also have practical implications for how the formulation is applied. HRT controls reactor volume in the present optimization problem. For an existing treatment plant, installed reactor volumes should instead be fixed and applicable effluent limits should be imposed explicitly. Mechanistic evaluation should also remain part of the decision process when surrogate prediction errors could affect compliance or operating choices. In this role, the plant-wide surrogate can reduce the computational work required to explore operating alternatives while the mechanistic model provides the final assessment of selected decisions.

Future work should evaluate the method across broader plant configurations, influent conditions, operating priorities, and treatment constraints. Additional starting points can be examined to determine the sensitivity of the selected controls to local search behavior. Alternative surrogate formulations and physical constraints could improve nutrient and solids responses while retaining the computational advantages of surrogate evaluation. Evaluation on independently specified cases would provide stronger evidence of performance beyond the conditions examined here. Further work on dynamic operating control would require assessing the effect of surrogate approximation on both treatment performance and the complete sequence of operating decisions.

CRediT authorship contribution statement

The sole author, Egberto F. Selerio Jr., contributed to conceptualization, methodology, software, validation, formal analysis, investigation, resources, data curation, writing—original draft, writing—review & editing, visualization, supervision, project administration, and funding acquisition.

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Data availability

The simulation data and numerical results supporting this study are available at <https://github.com/egbertoselerio1997/surrogate-optimization>.

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